ELi — The European Legislation Identifier

Good practices and guidelines

Author: **ELI Taskforce**

SECOND EDITION

Publications Office

More information on the European Union is available on the internet (http://europa.eu).

These methodology guidelines have been drafted with the support of ISA². ISA² is a EUR 131 million programme of the European Commission that develops digital solutions that enable interoperable cross-border and cross-sector public services for the benefit of public administrations, businesses and citizens across the EU. ISA² supports a large range of actions and solutions. The ISA² solutions can be used free of charge and are open source when related to IT.

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Luxembourg: Publications Office of the European Union, 2018

Print	ISBN 978-92-78-41433-7	doi:10.2830/737565	OA-04-17-130-EN-C
PDF	ISBN 978-92-78-41429-0	doi:10.2830/44907	OA-04-17-130-EN-N
ePUB	ISBN 978-92-78-41432-0	doi:10.2830/535911	OA-04-17-130-EN-E

Printed in Luxembourg

Printed on elemental chlorine-free bleached paper (ECF)

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Contents

Foreword

The present guide was created by the ELI Taskforce and is aimed at developers or project managers who want to implement ELI into their legal publishing systems.

The ELI Taskforce was set up in December 2012, under the auspices of the Council of the European Union Working Party on E-law, to study the future development of the ELI standard.

The taskforce aims to help Member States wishing to adopt ELI by sharing knowledge and expertise.

More information is available on the website: http://eurlex.europa.eu/eli.

Introduction

Purpose

The purpose of this document is to provide publishers of legal information with a step-by-step methodology on how to implement the European Legislation Identifier (ELI).

The current document is an updated edition of the ELI methodology guide that includes lessons learned from national ELI implementations and feedback received during hands-on workshops organised by the ELI Taskforce.

Structure

This document consists of three parts.

- **Part 1** provides an overview of ELI together with the main benefits derived from the implementation of ELI.
- **Part 2** provides a set of steps and good practices that publishers of legal information can apply in order to implement ELI. This part also provides an estimate of the resources needed to implement each of the three pillars of ELI.
- **Part 3** provides information on how ELI implementers can further enhance discoverability of legal information beyond the implementation of ELI.

Target audience

The target audience of this document is as follows.

- **Decision-makers.** Individuals who make decisions concerning the implementation of ELI in their countries. This group typically includes countries that have not yet implemented ELI but that are interested in learning more about the potential associated benefits and the experiences of those who have already implemented ELI.
- **Technology experts.** Individuals who want to learn how to implement ELI.
- **ELI implementers.** Those that who already implemented ELI in whole or in part and that are interested in the progress of other ELI implementers.
- **Legal professionals.** Individuals with a legal background that could be interested in learning more about URI templates in order to know how to better search for legal information.

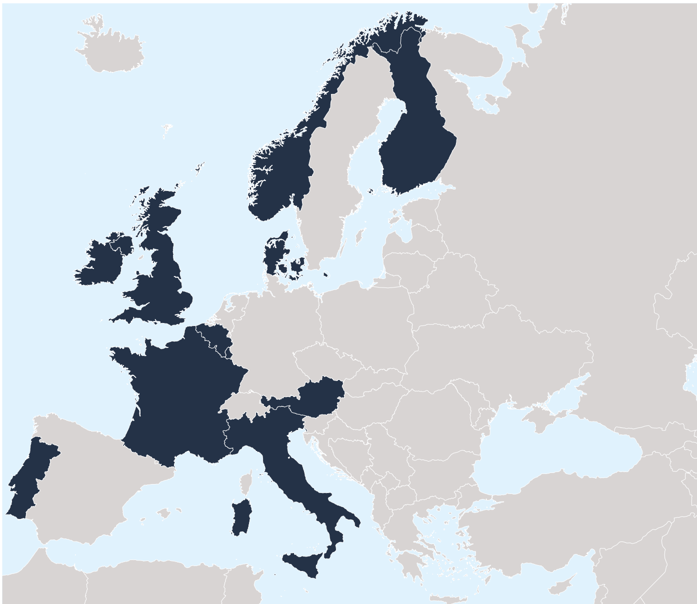

Figure 1 — Countries that have implemented ELI

Scope

Originally, the guidelines in this document were based on the findings gathered in a study carried out in the first quarter of 2015 with 11 Member States/European Free Trade Association (EFTA) countries (i.e. France, Germany, Ireland, Italy, Luxembourg, Malta, the Netherlands, Norway, Slovakia, Switzerland and the United Kingdom), along with the Publications Office of the European Union (OP).

In the course of 2015, 2016 and 2017, ELI workshops were delivered to Denmark, France, Ireland, Italy, Albania, Luxembourg, Malta, Finland, Austria, Serbia, Latvia, Spain, Portugal, Hungary and Croatia. The objective of the workshops was to provide technical consultancy to support the implementation of ELI in existing publishing systems.

Thanks to the feedback received in the context of these workshops, the methodology guidelines were enriched and refined.

PART 1:

An introduction to the European Legislation Identifier

Background

The conclusions of the Council of the European Union inviting the introduction of ELI explain that: '… a European area of freedom, security and justice in which judicial cooperation can take place requires not only knowledge of European law, but also mutual knowledge of the legal systems of other Member States, including national legislation' ([1]).

The exchange of legal information is key in this regard, but despite the increased availability of documents in electronic format, the Council conclusions report that the exchange of legal information originating from regional and national authorities at the European level is far from optimal, due to existing differences between national legal systems, including the technical systems used to store and display legislation through their websites.

To overcome such obstacles and improve interoperability between legal systems, the ELI Council conclusions invite the Member States to use identification of legislation and metadata properties describing each legal resource, and render the ELI metadata machine-reusable to reference national legislation in official journals and legal gazettes, so as to enable effective, user-friendly and faster search and exchange of legal information.

This is in line with the European Union's commitment to open up legislation as part of the implementation of the G8 Open Data Charter ([2]), which aims to promote, among other things, transparency and government accountability. It also meets the recommendations included in the European interoperability framework ([3]) on how to improve interoperability within the EU and across Member State borders and sectors, and also the directive on better regulation ([4]) on access to, and interconnection of, electronic communications networks and associated facilities.

By making legislation available on the web in a structured way, it will be easier to find, share and reuse legislation, as prescribed by the public sector information directive ([5]). Allowing the smart reuse of legal data and creating opportunities for the development of new services, ELI contributes to the development of the digital single market ([6]). Finally, ELI aims to promote the access and exchange of legal information within and across

([1]) OJ C 441, 22.12.2017, pp. 8-12 (http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52017XG1222(02)).

([2]) *EU implementation of the G8 Open Data Charter* (https://ec.europa.eu/digital-agenda/en/news/eu-implementation-g8-open-data-charter).

([3]) COM (2010) 744 final, Annex 2, Brussels, 16.12.2010.

([4]) Directive 2009/140/EC (http://eur-lex.europa.eu/legal-content/en/ALL/?uri=CELEX:32009L0140).

([5]) Directive 2003/98/EC (http://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32003L0098) and amending Directive 2013/37/EU (http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:32013L0037).

([6]) https://ec.europa.eu/priorities/digital-single-market_en borders, thereby contributing to the development of the common area of freedom, security and justice.

The ELI initiative has been supported since 2014 by the European Commission's 'Interoperability solutions for European public administrations' (ISA) programme. Since 1 January 2016 work on ELI has been pursued with the support of the ISA^2 programme (1) as action 2016.08 (2).

Origins

First established in the context of the European Forum of Official Gazettes (3), ELI has been further supported by the subgroup mandated by the Council of the European Union in the framework of the Working Party on E-law (4).

ELI stems from the acknowledgment that the World Wide Web defines a new paradigm for legal information access, sharing and enrichment.

In this context, consumers and publishers of legal information are faced with opportunities and challenges. On the one hand, there are new opportunities to access legal information in a more interconnected and interoperable way across legal information systems; on the other hand, there is a need to maintain the flexibility of each national legal system. Keeping this in mind, ELI is based on a progressive approach entailing the implementation of the following elements:

- identification of legislation via HTTP URIs;
- description of legal resources through metadata;
- rendering the ELI metadata machine-reusable through the serialisation of the metadata in compliance with the ELI ontology.

These three elements are commonly referred to as the three pillars of ELI.

Objectives

The purpose of ELI is to facilitate the access, sharing and interconnection of legal information published through national, European and global legal information systems. ELI aims primarily to facilitate the search, exchange and interconnection of legal information through national and European IT systems.

(1) http://ec.europa.eu/isa/isa2/index_en.htm

(2) *ISA^2 work programme*, 2018, p. 407 (https://ec.europa.eu/isa2/sites/isa/files/docs/pages/isa2_wp_2018_detailed_descriptions_part_1.pdf).

(3) https://publications.europa.eu/en/web/forum_official_gazettes

(4) http://www.consilium.europa.eu/en/council-eu/preparatory-bodies/working-party-e-law

The European Legislation Identifier is used to:

1. identify legislation with a unique identifier that is recognisable, readable and understandable by both humans and computers, and is compatible with existing technological standards;
2. describe legislation with a set of machine-readable metadata elements in compliance with a recommended ontology.

According to the Council of the European Union conclusions:

'ELI shall guarantee cost-effective public access to reliable and up-to-date legislation and is subject to voluntary and gradual introduction. To this end:

a) ELI creates a unique identifier for the legislation, which is readable by both humans and computers, and which is compatible with existing technological standards ("ELI pillar 1");

b) ELI proposes a set of metadata elements to describe legislation in compliance with a reference ontology ("ELI pillar 2");

c) ELI permits a greater and faster exchange of data: when this metadata is embedded in the respective webpages of the Official Journals and Legal Gazettes or legal information systems, information can be exchanged automatically and efficiently, thanks to the benefits from the emerging architecture of the semantic web, which enables information to be directly processed by computers and humans alike ("ELI pillar 3").

ELI gives the Member States and the European Union a flexible, self-documenting, consistent and unique way to reference legislation across different legal systems. ELI URIs are a stable means of uniquely identifying any legislative act throughout the European Union, while taking into account the specificities of national legal systems ([1]).

ELI takes into account not only the complexity and specificity of regional, national and European legislative systems, but also changes in legal resources (e.g. consolidations, repealed acts etc.). It is designed to work seamlessly on top of existing systems using structured data and can be taken forward by any national legislation publisher at European level and beyond at their own pace.

Apart from Member States, candidate countries, Lugano States ([2]) and others are encouraged to use the ELI-system.'

([1]) The European Case Law Identifier (ECLI), applicable on a voluntary basis, provides a European system for the identification of case-law. ELI identifies legislative texts which have different and more complex characteristics, and the two systems are complementary. The Council requested the introduction of the European Case Law Identifier and a minimum set of uniform metadata for case law by way of conclusions (OJ C 127, 29.4.2011, p. 1).

([2]) Iceland, Norway and Switzerland.

While legal information has been available online via national legal information systems for a long time, ELI enables the putting in place of a genuine network of legal information, available as open data and reusable by all.

To achieve these objectives ELI relies on the infrastructure of the World Wide Web and the technologies of the semantic web. ELI is based on an approach that is characterised by the progressive adoption by the publishers of legal information of the following elements.

1. **Pillar 1 on the identification of legislation:** URI templates at the European, national and regional levels based on a defined set of components.
2. **Pillar 2 on metadata properties describing each legal resource:** definition of a set of metadata and its expression in a formal ontology.
3. **Pillar 3 to render the ELI metadata machine-reusable:** integration of metadata into the legislative websites. In the annex to the Council conclusions it is suggested that the ELI metadata elements may be serialised in compliance with the World Wide Web Consortium (W3C) recommendation 'RDFa in XHTML: syntax and processing'.

The ELI Council conclusions do not specify as a technical requirement that the order of the pillars should be followed in order to implement ELI. However, the most common approach is to progressively implement ELI in line with the numbering of pillars.

ELI is based on the well-established model of 'Functional requirements for bibliographic records' (FRBR). FRBR distinguishes between the concepts of 'work' (intellectual or artistic creation), 'expression' (intellectual or artistic realisation of a work) and 'manifestation' (materialisation of one of the expressions of a work). Each time a 'work' is realised it takes the form of an 'expression'. The physical embodiment of an expression is a 'manifestation'. Figure 2 below illustrates how these concepts map to the ELI Council conclusions ([1]) as published in EUR-Lex. More specifically, the following can be stated.

- **LegalResource** is the conceptual entity that represents the intellectual content of the ELI Council conclusions, which is what the various language versions share. The intellectual content of the Council conclusions is realised by the text, which is expressed in various language versions.
- **LegalExpressions** are the language versions that are presented horizontally and express the intellectual content in actual text. The language versions are embodied by physical files.

([1]) OJ C 325, 26.10.2012.

- **Formats** are the physical files presented vertically under each language.

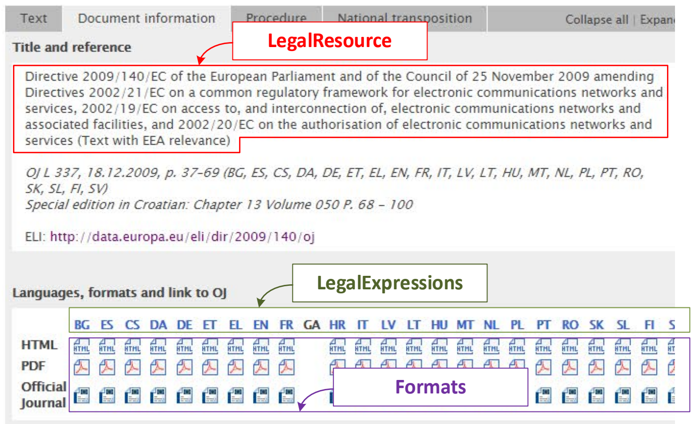

Figure 2 — Screenshot from the EUR-Lex website depicting the three conceptual entities of the ELI model

Governance

At the meeting of the Working Party on E-law on 20 December 2012, the chair of the working party proposed that an ELI Taskforce should be set up to further study and define the development of a unique identifier for national and EU legislation.

The ELI Taskforce was initially composed of France, Luxembourg (chair), the United Kingdom and the Publications Office of the European Union; Denmark, Finland, Ireland, Italy, Norway and Portugal joined the taskforce subsequently. In order to help Member States, candidate countries, EFTA countries and others to adopt ELI, the taskforce aims to share knowledge/expertise without imposing a strict schema on how ELI should be implemented and by taking into account national specificities. For this purpose a change-and-release management process is in place, and users can contact the Publications Office for any information they would like to receive and to provide feedback and comments on ELI [1].

In 2017 the Council of the European Union further expressed its support to the ELI initiative by creating an expert group exclusively dedicated to driving forward the European Legislation Identifier.

ELI is an action funded under the 'Interoperability solutions for public administrations, businesses and citizens' programme (ISA²). The Publications Office ensures the compliance of the activities conducted

(1) Contact: OP-ELI-MAILBOX@publications.europa.eu

with the objectives of the ISA² work programme and regularly reports to the Commission's ISA² coordination structure on the main achievements ([1]).

ELI benefits

The implementation of ELI brings a number of benefits for public administrations and publishers of legal information, as well as citizens at large.

Benefit 1: access to legislation

The large amounts of legislation, along with its diversity, can make access to it cumbersome. Thus, it is important that legislation be published in a way that allows people to use it easily and readily.

First of all the assignment of persistent identifiers as specified by ELI provides for a homogeneous mechanism to identify legal information on the web, making it easier for users to look for legal information within and across legal systems.

The predictable nature of the ELI identifiers, which are designed in such a way as to stay as close as possible to the way legislation is cited, allows users to easily compose URIs to refer to published legislation. What is more, the design patterns of the ELI-compliant HTTP URIs are conceived to assist users looking for legal information when they do not have all the necessary elements to identify the specific piece of legislation. In this case, the user will be directed towards the right information through a partial input, based on the available information.

ELI also facilitates access to legal information by citizens, along with potential reusers of legal information. It allows legislation to be referenced at different levels of granularity, thus enabling different uses of legal information, such as social media discussions about particular elements of legislation, references to legislation in research papers or developments of applications by businesses.

Furthermore, a common structure for metadata to describe legislation across national and European legislation services, as proposed by ELI, makes searching across the distributed collections easier. It also creates opportunities for aggregation services to harvest metadata from various sources to build indexes across collections.

Additionally, ELI promotes the access and reuse of legal information across Europe through multilingualism that is facilitated by the use of its ontology, multilingual controlled vocabularies and the alignment of controlled vocabularies with EU vocabularies.

([1]) https://ec.europa.eu/isa2/actions/facilitating-exchange-legislation-data-europe_en

Benefit 2: development of new services

The ever-evolving social, economic and technological environment in which publication services are operating enables smart reuse of data. The persistence of identifiers, the common specification of structured metadata and the publication format proposed by ELI give businesses the opportunity to develop innovative and sustainable added-value services on the basis of the published legal information.

Furthermore, the increased semantic interoperability in the context of legal information achieved through the adoption of ELI contributes to the development of the digital single market, which will lead to the modernisation of traditional industry, and bring estimated significant gains in terms of additional annual growth, along with a substantial boost for job creation in Europe.

Benefit 3: cost savings

Failure to link or share legislation within and between different public administrations results in inefficient public services and high information collection costs, increasing the burden on public administrations and citizens.

ELI promotes the publication of legislation in a structured and machine-readable way to promote linking and reuse of legal data. This also leads to cost savings for publishers, in terms of improved effectiveness of information flows and shorter time required to publish legislation.

The ELI approach aims to construct common building blocks for naming and citing legal information in a flexible way so as to respect each country's unique legislative and legal tradition. This flexibility — together with the fact that ELI can be implemented in a progressive way on top of existing IT solutions and databases, and that it is designed to work seamlessly on top of existing systems — makes it a cost-effective solution for publishers that want to make legal information available for reuse purposes.

Benefit 4: quality and reliability of data

The increased reuse of legislation triggers a growing demand to improve the quality of the published information. The structured publication of legal information based on the ELI specifications enables review and feedback from users, which can help increase the quality and reliability of information.

Furthermore, ELI allows better structuring of legal information in terms of internal links within a text and external links to other documents that are mentioned within a text (for example laws, decrees or EU legislation).

Quality and reliability of data is key in order to establish more effective, simplified and user-friendly services that are crucial for fostering the trust of businesses and citizens in digital services.

Benefit 5: transparency

There is an ever-growing demand for openness and transparency in modern societies. Increased transparency is an important objective across public administrations, as demonstrated in the context of the European Union, where the principles of openness and transparency feature in primary EU law.

By making legislation more accessible, ELI contributes to making it easier for citizens and watchdog organisations to be better informed and to monitor the work done by governments. Transparency guarantees greater legitimacy and accountability of the administration in a democratic system because citizens have the opportunity to better understand the considerations underpinning the law in order to exercise their democratic rights.

Benefit 6: interoperability

ELI is based on the architecture of the semantic web and linked open data, enabling greater and faster exchange of data through the automatic and efficient exchange of information. This is achieved through semantic interoperability. Semantic interoperability is a means to achieve greater efficiencies where there is a clear need and where there is clear demand from relevant users.

The ELI approach is based on the idea that in order to generate greater efficiencies it is necessary to reach a minimum level of agreement on the way legislation is identified and described. The ELI approach thus allows legislation to be structured and identified in a sufficiently uniform way, and at the same time takes into account the specificities of each national legal system by providing a flexible and progressive approach to its implementation.

ELI also provides a standard way to publish legislation as machine-readable open data, creating opportunities to automatically discover and re-use published legislation and to ensure legal reference certainty.

By allowing humans and machines to have improved access to legal information, ELI sets the basis for a web of legal information. Making legal information part of the web of data allows users to easily discover legal information across countries and to reference legal information more easily, as well as to improve their understanding of legal information thanks to the interconnections between different sources of information across domains.

PART 2:

ELI implementation methodology

Overview of the ELI implementation methodology

This section introduces the ELI implementation methodology, providing a step-by-step approach that publishers of legal information can follow to implement one or more ELI pillars. This is accompanied by a set of good practices that have been identified based on the experience of stakeholders that have already implemented one or more of the ELI pillars. The ELI implementation methodology consists of two parts, as depicted in Figure 3 below.

1. **Organisation and policy:** this part mainly addresses decision-makers and is composed of five main steps.
2. **Technical implementation:** this part addresses a more technical audience that is involved with the actual ELI implementation. It is divided according to the three ELI pillars, and for each pillar a number of steps have been outlined.

The ELI implementation methodology is generic enough to cover a large number of possible use cases. To be effective, the methodology should be tailored and adapted to match the use case of the implementer. Despite the order of pillars and their steps forming a coherent flow — i.e. (1) design and implement ELI identifiers for legislation, (2) design and implement metadata management and (3) publish legislation enriched with metadata — not all pillars and steps have to be implemented at once. Some may decide to limit their implementation only to using ELI identifiers for all their published legislation. Others may implement all three pillars at once for a limited part of their published legislation, and then gradually for all of it.

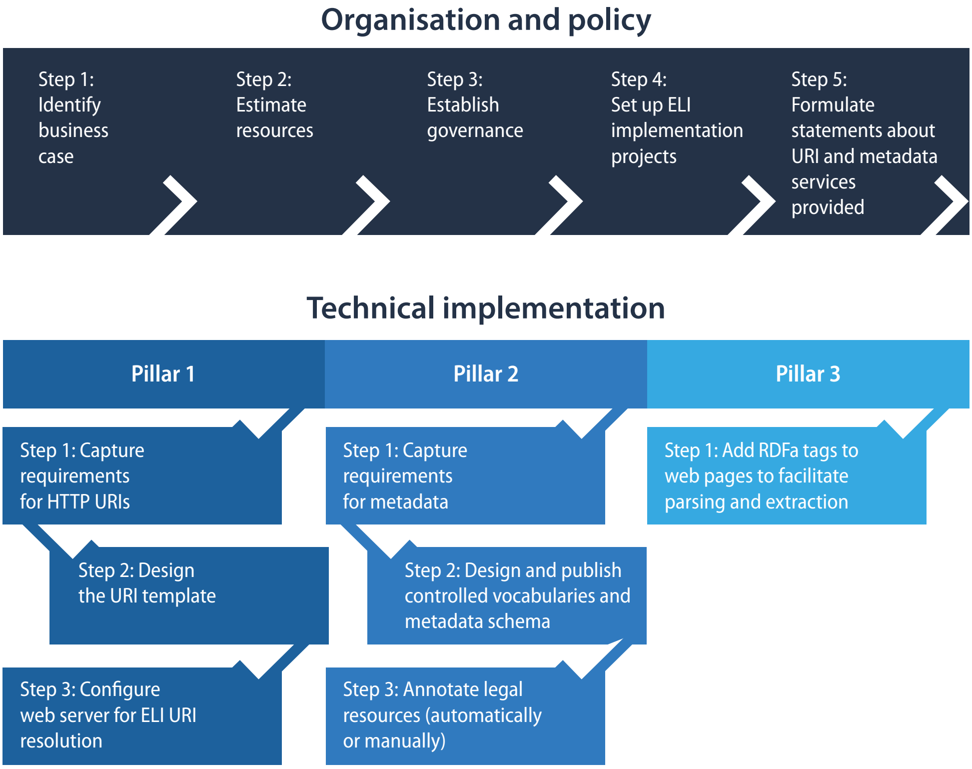

Figure 3 — Overview of the ELI implementation methodology

Organisation and policy

STEP 1: Identify the business case

Publishers of legal information should develop a business case to help decision-makers decide whether to implement ELI and the allocation of resources that are necessary to implement it. The business case should include the following elements.

- **An executive summary** that outlines the ELI project and contains the key considerations that will be discussed in the business case, including the timeline for business case implementation and completion, along with the projected benefits and costs.
- **The scope** defining the coverage of ELI implementation from both a chronological and a type(s) of collection perspective. With regard to the chronological aspect, publishers of legal information need to decide whether ELI will be implemented in relation to all legislation that has ever existed or to a subpart, such as only newly adopted legislation. This decision depends significantly on what legislation is available online or is planned to be made available online.

  With regard to the type(s) of collection, publishers of legal information may decide that ELI will be applied only to legislation or to a broader range of legal information. For example, on the one hand, France decided to include summaries of legislation within the scope of ELI because this collection is regularly consulted by users to get a better understanding of legislation; on the other hand, the Publications Office decided to limit the scope of ELI implementation to the EU legislation published in the Official Journal L series.
- **A problem statement** highlighting the issue or goal that the business case for the implementation of ELI aims to solve. The 'ELI benefits' presented in Part 1 of this guide can provide guidance on common issues and objectives that ELI aims to address. An example of a concrete problem statement entails the need to establish clear links between an EU directive and related national implementing measures. The issue relates to the fact that it is not always easy to establish which national provision transposes which EU directive, and even if the relationship is identified there is a need to ensure that such information is kept up to date. In the past, public administrations used static tables of correspondence that were manually maintained. Such an exercise was not only extremely cumbersome, it also led to information that was not always accurate and up to date.

- **A proposed solution** that describes the project and options available to solve the business problem, explaining how the implementation of ELI addresses and resolves the problem statement. For example, ELI provides a useful means to improve the links between EU directives and national implementing measures as it promotes semantic interoperability between legal information systems across countries.
- **An estimate** of resources that indicates what is needed to implement ELI, including the allocation of resources, a timeline for implementation and the resources needed. For guidance on how to define the resources necessary for ELI implementation the reader is invited to move to the next step, 'Estimate resources'.

Good practice 1:
Building on the knowledge, experience and tools of others (support by the ELI Taskforce)

Problem statement

When approaching ELI implementation, publishers of legal information may be inclined to work in isolation, without looking for existing experience and know-how. This leads to inefficiencies as time and money may be spent investigating aspects that are already known inside other organisations that have already implemented ELI.

Recommendation

Publishers of legal information can learn from the experience of those who have already implemented ELI and can ask for help and assistance from the ELI Taskforce. The references of ELI coordinators and ELI Taskforce members is made available on the EUR-Lex website ([1]). EUR-Lex also provides a list of tools that can help in the implementation of ELI ([2]). Additionally, the Publications Office's Metadata Registry provides references to the ELI URI template components and ELI ontology, as well as to the serialisation of ELI metadata elements ([3]).

Building on shared knowledge allows publishers of legal information to anticipate risks and avoid mistakes made by others in the past, saving time and resources. Learning from the findings of legal analysis already carried out by other publishers, and from technical implementation, helps avoid reinventing the wheel and optimises the use of resources.

For instance, some countries greatly benefited from the support provided by the ELI Taskforce, in terms of both technical support and more general background knowledge, in order to acquire a better understanding of how ELI works and the benefits derived from its implementation.

DOs

- Learn from the experience of those who have already implemented any of the pillars of ELI
- Share lessons learned while implementing ELI and provide feedback to the ELI Taskforce

DON'Ts

- Work in isolation
- Reinvent the wheel

BENEFITS

Quality and reliability
Avoids common pitfalls

Development of new services
Optimisation of resources and better focus on aspects to promote the development of added-value services

([1]) http://eur-lex.europa.eu/eli-register/implementation.html
([2]) http://eur-lex.europa.eu/
([3]) http://publications.europa.eu/mdr/eli/index.html

STEP 2: Estimate resources

When estimating the resources to implement ELI, publishers of legal information should take into account the fact that the effort required for ELI implementation varies depending on:

- each ELI pillar; and
- the degree of digitalisation of the existing publication process for legal information.

To help publishers carry out such an analysis, this document provides the following overview of the resources that should be allocated depending on whether:

1. a publisher of legal information needs to start implementation from the very beginning (**Scenario 1**); or
2. a publisher of legal information is already well advanced (**Scenario 2**).

For each scenario, it is possible to identify:

- the resources estimate associated with each of the implementation steps, which are further explained in the remainder of this document; and
- relevant professional profiles.

The following two key professional profiles are considered relevant.

- **Knowledge information officer:** should have a good understanding of information technologies in general and the semantic web in particular, and expert knowledge of legal information. Experience in working with multiple metadata standards, frameworks and controlled vocabularies, familiarity with XML and XML schema languages together with knowledge of XPath, XSLT or XQuery are recommended. Furthermore, experience in creating and implementing metadata schemas and taxonomies and controlled vocabularies, together with gathering user requirements, is a plus.
- **Semantic web programmer/developer:** should be proficient in software development in general, and have basic experience with semantic web technologies. Skills necessary for a web programmer/developer are experience in transforming data format using XSLT and adding RDFa or JSON metadata into HTML pages.

Figures 4, 5 and 6 below provide an overview of the estimated resources for the implementation of each ELI pillar. Implementation efforts are expressed in full-time equivalent (FTE), which is a unit to indicate the workload of an employed person.

Figure 4 — Resources allocation Pillar 1

Activity	Range	Person days	Profile
Scenario (1) No metadata is available			
Step 1: Capture requirements for metadata	*min 5 – max 15*		Knowledge information officer
Step 2: Design and publish metadata schema	*min 15 – max 30*		Semantic web programmer/developer
Step 3: Develop metadata publishing routines	*min 10 – max 20*		Knowledge information officer
Step 4: Deploy legislation metadata	*min 20 – max 40*		Semantic web programmer/developer
	person days	0-10 / 11-20 / 21-30 / 31-40	

Activity	Range	Person days	Profile
Scenario (2) Required metadata is available, but not aligned to ELI			
Step 2: Design metadata schema	*min 2 – max 5*		Knowledge information officer
Step 4: Deploy legislation metadata	*min 3 – max 10*		Semantic web programmer/developer
	person days	0-10 / 11-20 / 21-30 / 31-40	

Legend:
- Best-case scenario
- Average scenario
- Worst-case scenario

Figure 5 — Resources allocation Pillar 2

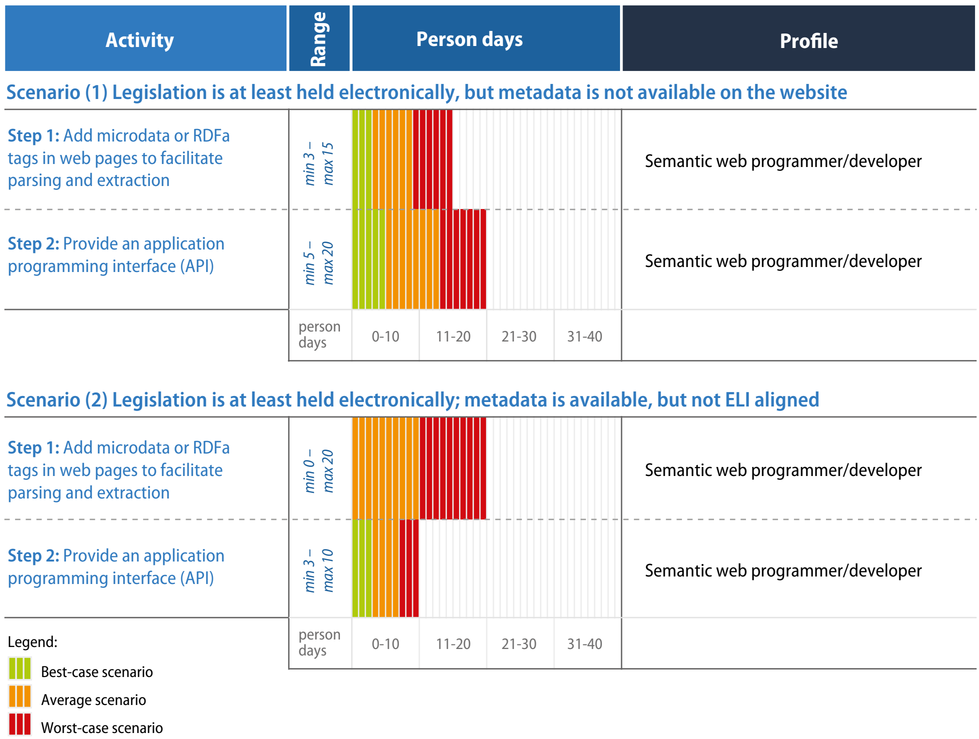

Figure 6 — Resources allocation Pillar 3

Good practice 2: Estimating implementation costs

Problem statement

In order to decide whether to implement ELI, it is fundamental to provide an accurate estimate of implementation costs. The development of semantic web solutions such as ELI necessitates publishers of legal information to take into account not only the technical developments required to implement ELI but also other associated costs.

Recommendation

To secure the appropriate allocation of funds, publishers of legal information should take into account three main cost drivers:

- product, including hardware and software;
- process, including organisational aspects;
- personnel, taking into account the need for different profiles.

To ensure accurate estimates, publishers of legal information should identify constraints so as to ensure that the estimates are meaningful.

The cost estimate should take into account not only the costs encountered during the development of each ELI pillar, but also the maintenance of the service provided. Publishers of legal information can rely on a variety of approaches to carry out their cost estimate including:

1. consulting the estimate of resources provided in this document;
2. comparing costs encountered by other publishers of legal information when implementing ELI or similar projects;
3. involving technical experts to estimate the necessary effort.

BENEFITS

Quality and reliability

Support to the effective implementation of ELI with appropriate allocation of resources

DOs

- Estimate effort, for example using full-time equivalent (FTE), which is a unit that indicates the workload of an employed person in a way that makes workloads across various contexts comparable
- Estimate cost for hardware and software
- Estimate cost to manage risks

DON'Ts

- Estimate implementation effort without consulting technical experts
- Underestimate coordination efforts

STEP 3: Set up a governance structure

Once the decision to implement ELI has been taken, publishers of legal information should ensure that all stakeholders involved in the implementation project have a clear understanding of the decision-making process. For this purpose it is recommended that a governance structure be set up that, for example, could be composed of: a steering committee, providing direction for implementation and review of the progress of ELI implementation; a governance committee, in charge of managing the operational team(s); and an operational team (or several such teams), mainly composed of technical staff, responsible for the implementation and maintenance of ELI.

When setting up the governance structure, publishers of legal information should take into account the ELI Council conclusions, which establish, in Title VI, that Member States should appoint a central organisation as the national coordination point for ELI. More specifically, the annex states the following.

'On national implementation

1.1. The national ELI coordinator

1. Each country using ELI must appoint a single national ELI-coordinator.

2. The national ELI-coordinator is responsible for:

   a) reporting on the ELI implementation status;

   b) sharing and reporting on the applicable URI template(s);

   c) sharing and reporting available metadata and its relationship to the ELI metadata schema (if applicable);

   d) providing the above information to the ELI TF and the Expert Group for publishing on the ELI-Register website.'

Setting up a national coordination point facilitates the adoption of common policies relevant in the context of the ELI implementation, such as policies on persistent identifiers and metadata. Establishing a common policy, while not obligatory, can help in ensuring that relevant technical and organisational aspects are identified and tackled in a coordinated manner before any project is started. The references of all national ELI coordinators are available on the ELI website (EUR-Lex).

Once the organisational structure is established, publishers of legal information can then define change-and-release management procedures and quality control procedures. The ELI Taskforce can provide guidance in this regard, if their assistance is requested.

Good practice 3:
Setting up a central organisation as a national coordination point for ELI implementation

Problem statement

ELI implementation requires carrying out a number of tasks which, depending on the existing administrative structure, might be assigned to different entities. Such a decentralised organisation, however, might lead to a number of drawbacks, most notably:

- inconsistencies in formulating coherent URI and metadata policies;
- increased costs for ELI implementation and maintenance.

Recommendation

In line with Title VI and the annex to the ELI Council conclusions, publishers of legal information should appoint a national ELI coordinator responsible for:

' a) reporting on the ELI implementation status;

b) sharing and reporting on the applicable URI template(s);

c) sharing and reporting available metadata and its relationship to the ELI metadata schema (if applicable);

d) providing the above information to the ELI TF and the Expert Group for publishing on the ELI-Register website.'

The national coordination points for ELI implementation are available on the ELI website.

DOs

- Follow the ELI Council conclusions and appoint a national coordinator for ELI implementation

DON'Ts

- Allow a decentralised ELI implementation at national level

BENEFITS

Interoperability
Coherent URI and metadata policies implemented at national level

Quality and reliability
Increased opportunities to align ELI implementation efforts at national level

STEP 4: Set up an ELI implementation project

ELI can be implemented progressively. This means that publishers do not have to go through the full implementation of the three pillars at once. Instead, it is possible to take a step-by-step approach. For example, Denmark, France, Italy, Luxembourg, Finland and the Publications Office implemented the three ELI pillars all at once, but reduced the scope of the implementation to certain content types (i.e. statutory instruments, basic acts excluding consolidated texts, or Official Journal issues). Following such an implementation approach, the next steps should iteratively increase the scope of work to include the remaining content types. Another example of structuring an ELI implementation project comes from the United Kingdom, which, at the time of drafting this methodology, had implemented the first of the three pillars. The United Kingdom provides ELI-compliant URI patterns. There is work in progress to convert and publish current metadata to ELI equivalents. Similarly, Ireland first implemented Pillar 1, and waited for the implementation of its new legal portal before implementing Pillars 2 and 3.

ELI implementation can be split into multiple phases to increase control over the implementation. Each phase could start out small in scope by including only a certain type of content (e.g. statutory instruments), and could iteratively scale up by including the remaining types of content (e.g. consolidated texts, Official Journal issues) as the implementation advances and the expected results become tangible.

Some countries that already have well-structured legislation may implement all of the pillars of ELI in the first phase, as a prototype with little or no impact on the publication processes. In the next phase the result of the prototype could be integrated within the publication processes. Other countries with very little metadata could include Pillars 1 and 2 in the first phase to define a minimum set of metadata, design ELI URI patterns and enrich legislation with metadata. In the next phase the implementation would continue with Pillar 3, to annotate and publish legal information.

The fact that publishers of legal information decide at a certain point to limit ELI implementation to one or two pillars does not exclude the possibility of implementing the third one later on. Keeping this flexibility in mind, the key steps that publishers of legal information need to take into account when setting up the ELI implementation project are as follows.

1. **Define priorities** based on user requirements (cf. 'Organisation and policy', Step 1) and policy agendas. Since ELI can be implemented following a step-by-step approach, publishers of legal information can prioritise deliverables between 'must have' and 'nice to have'.

2. **Set up a team** including a project manager, a user representative, experts in information management and the semantic web and additional stakeholders who are impacted by the implementation of ELI.

3. **Choose an implementation approach** that emphasises the use of iterative developments whereby regular releases are envisaged (for example every 3 months) that allow ELI implementation to be built and refined based on user feedback and use cases.

4. **Build the project timeline** taking into account the time required to design, build and deploy the ELI pillars. By following the iterative implementation approach the timeline will consist of short iterations of the design, construction and deployment phases for each pillar.

5. **Implement the ELI pillars** in line with the developed timeline and following the implementation steps provided in this document (cf. Pillar 1, 'How to put in place legislation identifiers based on ELI HTTP URIs', Pillar 2 'How to design an ELI metadata schema' and Pillar 3 'How to render ELI metadata machine-reusable').

6. **Encourage use and raise awareness** by providing relevant documentation and presenting the service via workshops, webinars and conferences.

Setting up the ELI implementation project will help to:

- identify high-level implementation options: typically this would entail an analysis of implementing Pillar 1 only, the first two pillars or all three pillars;
- understand whether to go ahead with the implementation and with which pillars;
- define the efforts required for ELI implementation and for the deadlines to be met;
- target the main functionalities to be implemented while adopting ELI.

Good practice 4:
Structuring your implementation project

Problem statement

Each ELI pillar can be implemented progressively. Therefore, publishers of legal information have flexibility in deciding how to go about the implementation of ELI. However, the great flexibility offered by ELI may be challenged by the absence of a well-structured project, as it can be difficult to identify the right scope and activities.

Recommendation

In order to ensure the successful implementation of ELI, publishers of legal information need to have a structured methodology for all the execution phases of the project. The methodology described in this document can be used to structure the work plan. Overall, key steps to be taken into account include:

- gathering requirements;
- analysing existing publication processes;
- designing the solution;
- testing against real data;
- implementation.

Publishers of legal information can decide upon the scope of implementation based on content types (i.e. statutory instruments, basic acts excluding consolidated texts, or Official Journal issues). For each content type, the implementation of the three ELI pillars can be broken down as follows.

- **Pillar 1 (HTTP URIs):** 1. capture requirements for ELI HTTP URIs; 2. design the URI template; 3. configure the web server for ELI URI resolution.
- **Pillar 2 (metadata):** 1. capture requirements for metadata; 2. design and publish the metadata schema; 3. develop metadata extraction routines; 4. deploy legislation metadata.
- **Pillar 3 (rendering ELI metadata machine-reusable):** 1. add RDfa markup to web pages to facilitate parsing and extraction.

BENEFITS

Quality and reliability

Project management in a control environment helps to deliver within agreed scope, budget, time and quality

DOs

- Build a proof of concept to demonstrate ELI implementation feasibility
- Break down the large-scale implementation into phases
- Take into account risks

DON'Ts

- Initiate project before carrying out assessment and planning
- Postpone scope definition

STEP 5: Formulate statements about URI and metadata services provided

Once publishers of legal information decide to implement one or more ELI pillars it is advisable to formulate a statement indicating the key aspects offered by the service provided. Important elements that the statement should contain are as follows.

- **Governance:** the national coordination point for ELI should be described as being the authoritative source of information for the implementation of ELI.
- **ELI URI template(s):** the statement should provide information about the URI template(s) for each type of published document (i.e. statutory instrument, consolidated texts or Official Journal issues). The statement could be published on the ELI website.
- **Metadata:** the statement should also refer to the metadata for each type of published document if the ELI implementation also includes the second pillar. Such information should also provide information on existing relationships supported by the ELI metadata, along with mappings with other standards (cf. Pillar 2, Steps 2, 3 and 4). Furthermore, information about the quality of metadata could be included, as this is also important in terms of liability for the publishers of legal information.

In general, publishers of legal information are encouraged to say what level of commitment they want to set up, without having to necessarily quantify it.

Finally, the statement should also address technical and non-technical profiles and should be published on the ELI website to raise awareness and promote the reuse of legal information.

Examples of statements about ELI URI can be found by following the links below.

PARTICIPANT	REFERENCE TO DOCUMENTATION
France	http://www.eli.fr/en/constructionURI.html
Ireland	http://www.irishstatutebook.ie/pdf/ELI_URI_schema.pdf
Publications Office	http://publications.europa.eu/mdr/eli/documentation/uri_templates.html
United Kingdom	http://www.legislation.gov.uk/developer/uris
Denmark	https://www.retsinformation.dk/eli/

Good practice 5:
Writing a policy document providing guarantees for long-term persistence

Problem statement

The absence of sufficient guarantees for the persistence of URIs and metadata when implementing ELI can have a negative impact on the level of trust among potential consumers of legislation. Reduced trust may, in turn, result in limited use of the legal information published, therefore limiting the potential for the creation of added-value services.

Recommendation

By drafting a policy document that provides sufficient guarantees with regard to the persistence and quality of the information provided, publishers increase confidence in relying on HTTP URIs and metadata to access and reuse legal information. Such a policy document should address the following.

1. Needs: for example, the need for stable URIs or the need to avoid duplication of identifiers for the same resource.

2. Governance structure: there should be a short description of how the management of URIs and metadata is performed and who is responsible for the proper maintenance of the publication service.

3. Technical aspects: the policy document should make reference to the technical aspects of the URI policy and metadata policy. The policy document should be published on the ELI website and promoted by the publishers of legal information.

BENEFITS

Access to legislation
Clear understanding of access and reuse requirements

Interoperability
Promotion of greater interoperability thanks to clear documentation

Quality and reliability
Increased level of commitment to provide a quality data service

DOs

- Take into account the persistence requirement when designing ELI URIs
- Write and publish a policy document for persistent ELI URIs and metadata on a dedicated ELI website
- Describe how persistence will be ensured if the national coordination point changes
- Consult policy documents of those who have already implemented ELI

DON'Ts

- Ignore the needs of consumers of legislation to have guarantees of persistence
- Take for granted the trust of consumers of legislation in service levels

Pillar 1:
How to put in place legislation identifiers based on ELI HTTP URIs

STEP 1: Capture requirements for HTTP URIs

The first step that publishers of legal information should take when approaching the implementation of the first ELI pillar (i.e. HTTP URIs) is to get a detailed understanding of user needs and, if necessary, update the business case (cf. 'Organisation and policy', Step 1). Many techniques are available for gathering user requirements. In order to gain a complete overview of the requirements that are relevant for the implementation of ELI HTTP URIs, publishers of legal information might consider the following approaches.

- **Interviews:** publishers of legal information should gather detailed information on what type of tasks users carry out with legal information, and understand how they cite legal information and what type of information they expect to find online. Interviews are very useful in this regard because they allow an in-depth analysis to be carried out and new requirements uncovered.
- **Questionnaires:** questionnaires are useful when publishers of legal information need to gather a large number of requirements from a wide audience, including users in remote locations that cannot be interviewed individually. It is advisable to complement the results gathered via questionnaires with more in-depth approaches.
- **Prototyping:** this technique allows requirements to be gathered that will help publishers of legal information to build the initial version of the HTTP URI. The development of a prototype on HTTP URIs will help users provide feedback that will be used to refine the HTTP URI implementation. This process should continue until the HTTP URI template meets the critical mass of user needs.
- **Use cases:** use cases are useful to describe how and for what purpose HTTP URIs for each type of content are used. Examples of use cases for HTTP URIs might include: consultation of legislation as it stood at a specific point in time; access to the consolidated version of specific piece of legislation; references to legal information in social media.

On top of gathering user requirements, publishers of legal information should take into account needs that are more related to the characteristics of legal information and to the environment where the HTTP URIs will be developed, such as the following.

- Whether there is a **single authority** that attributes the identifiers or whether the attribution of identifiers is done in a decentralised manner. For example, in a federal system like Germany, since there are 16 *Bundesländer* that promulgate legislation, the attribution of identifiers is decentralised.
- The fact that legal information might not all have an identifier to be used to **uniquely identify** legal information, which means that

alternative options should be considered. This is the case for Luxembourg where, for example the following regulation:

Règlement grand-ducal du 2 février 2015 portant organisation de la Conférence nationale des élèves

is identified uniquely with an ELI-compliant HTTP URI by adding a sequential number (cf. 'n7'), as shown below

http://eli.legilux.public.lu/eli/etat/leg/rgd/2015/02/02/n7 ([1]).

Another example can be seen in Norway. Before Lovdata ([2]) existed not all regulations had numbers. As a result, when Lovdata came into existence it introduced numbers to legislation in a systematic way. Where numbers did not exist (i.e. for regulations and statutes published before 1982) Lovdata retroactively assigned them, and from 1982 all regulations got an official number published in the Legal Gazette.

- The fact that legislation may be published in different websites, with different domain names. For example, in Italy, basic acts are published on the Gazzetta Ufficiale website, while consolidations can be browsed on the Normattiva website. Domain-agnostic URIs (e.g. http://data.country.eu/eli/, etc.) may help in this situation.

- The need to identify whether legislation is published as an individual resource, or in the context of an official journal, or both. For example, Italy provides ELI for both of the following.

    - Official journal:
      http://www.gazzettaufficiale.it/eli/gu/2015/1/13/9/sg/html.

    - Individual resource:
      http://www.gazzettaufficiale.it/eli/Decreto/2013/65/art4.

  For information on the different URI template components, please see the Italy page of the ELI website: http://eur-lex.europa.eu/eli-register/italy.html.

- Take into account when **consolidation** takes place (when applicable). For example, France consolidates legislation immediately after the amending act has been adopted and has put in place a specific database for it, which is called 'LEGI'. Ireland instead consolidates legislation on a case-by-case basis, and mostly for large acts.

([1]) The elements of the ELI templates are explained here:
etat = jurisdiction/agent, country-wide
leg = type, legislation
rgd = subtype, règlement grand ducal
2.2.2015 = date of signature
n7 = sequence number, distinguishing legislation signed on the same day

([2]) Lovdata is a foundation which publishes judicial information on Norway (https://lovdata.no/info/information_in_english).

- The existence of **more than one calendar.** This is the case for Ireland and the United Kingdom where, prior to 1922 and 1963 respectively, the regnal years were used to identify the year that legislation was published. To overcome such a challenge, these countries decided to redirect URIs with a calendar year before these dates to a URI based on the Gregorian calendar. Although such an approach worked for the majority of the cases, there were a small number of items of legislation held on the United Kingdom website where this redirection could not have been done unambiguously. For such cases a list of candidates is given in response to the user, who then has to choose the right legislation.
- The existence of **legacy identifiers.** It may be that when publishers of legal information implement ELI they already assign identifiers to legal information. When this is the case these identifiers can continue to be supported. For example, in the past Ireland went through two iterations of URI patterns. The first implemented pattern was not user friendly and it had to be replaced around 2004. The second URI pattern was a user-friendly pattern that had some resemblance to the current ELI URI schema. As there are still a large number of websites that link to the eISB (the electronic Irish Statute Book) using the old pattern, redirection is planned from the old to the new HTTP ELI pattern.

STEP 2: Design the URI template

The ELI Council conclusions provide the following components that publishers of legal information can use to design HTTP URIs for each type of content published.

/eli/{jurisdiction}/{agent}/{sub-agent}/{year}/{month}/{day}/{type}/{natural identifier}/{level 1…}/{point in time}/{version}/{language}

When designing ELI URI schemas it is important to make a distinction between the (abstract) legal resources and their realisations. In this way it will be possible to identify a reference to legislation independently of the version, language, representation format, etc.

For example, URI http://legilux.public.lu/eli/etat/leg/loi/2011/04/08/n2 identifies the abstract basic act 'Loi 1974 04 16'. When Legilux ([1]) receives a request for this kind of resource, it returns:

- the Legilux page of the latest consolidated version of an act;
- if no consolidated version exists, the Legilux page of the base act (identified by http://legilux.public.lu/eli/etat/leg/loi/2011/04/08/n2/jo);
- and always includes the not-yet-integrated 'rectificatifs' (amendments), if any (e.g. http://legilux.public.lu/eli/etat/rgd/2011/04/08/n2/republication/20140331/jo).

The picture below shows examples of ELI URI templates of an abstract legal resource, a basic act, modifying acts and their consolidations.

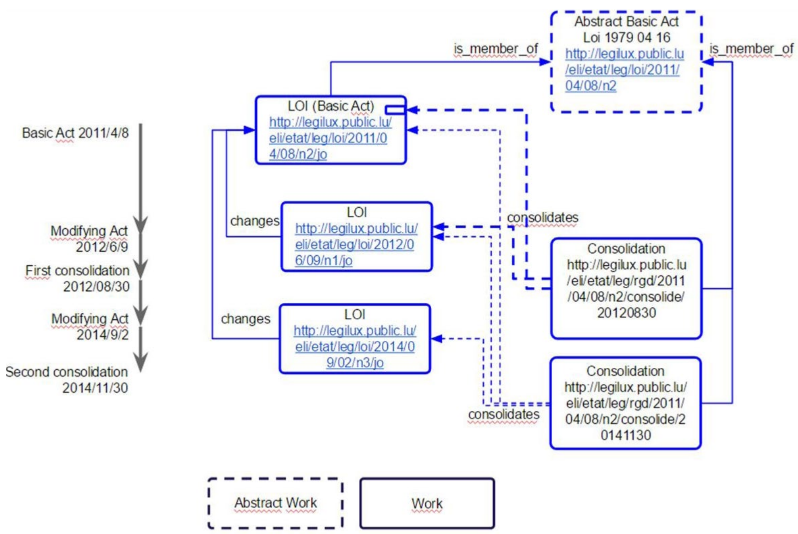

Figure 7 — Example of URIs for an abstract legal resource, basic and modifying acts and consolidations

([1]) Journal Officiel du Grand-Duché de Luxembourg: www.legilux.public.lu.

Examples of identifiers close to the way legislation is cited

LEGAL INSTRUMENT	IDENTIFIER FAR FROM THE WAY LEGISLATION IS CITED	IDENTIFIER CLOSE TO THE WAY LEGISLATION IS CITED (ELI HTTP URI)
EU: Directive 2003/98/EC of the European Parliament and of the Council of 17 November 2003 on the reuse of public sector information	32003L0098	http://eur-lex.europa.eu/eli/dir/2003/98/oj
Italy: Decreto 27 febbraio 2013, n. 65 Regolamento, di cui all'articolo 16, comma 1 del decreto legislativo 1° giugno 2011, n. 93, per la redazione del Piano decennale di sviluppo delle reti di trasporto di gas naturale	13G00106	www.gazzettaufficiale.it/eli/Decreto/2013/65

In addition to design, ELI URI schemas usually match the types of legal resources, but it is also important to take into account how people will use URIs. People should find identifiers of legal information that are easy to use, thus the closer identifiers are to the way the legislation is cited the easier it is to understand them, remember them and use them. For illustrative purposes, two examples are provided below. The first identifier is far from the way legislation is cited, the second one (ELI HTTP URI) is close to the way legislation is cited.

As the examples above demonstrate, it is easier to remember and re-use an identifier that is close to the way legislation is cited compared to a number with a more complex structure that is less intuitive.

When designing the ELI URI template(s), publishers of legal information should follow the following steps:

- identify how legal information is cited (Good practice 6);
- ensure that each legal resource is identified uniquely (Good practice 7);
- take into account the type of content to be published (e.g. statutory instruments vs acts, basic acts vs consolidated texts, official journals) and the concepts of the ELI ontology (legal resource, legal expression and format) (Good practice 8);
- identify subparts of legal information (Good practice 9);
- follow good practices for HTTP URIs (Good practice 10); and
- test URI templates (Good practice 11).

Publishers of legal information can find a number of examples of ELI HTTP URIs on the ELI website. An example is the HTTP URI template implemented in Luxembourg, shown together with a concrete example in the table below.

COUNTRY/LEGAL INSTRUMENT	ELI TEMPLATE/EXAMPLE
Luxembourg	/eli/etat/leg/{type}/{année}/{mois}/{jour}/{id}
Règlement grand-ducal du 2 février 2015 portant organisation de la Conférence nationale des élèves	http://eli.legilux.public.lu/eli/etat/leg/rgd/2015/02/02/n7
France	/eli/{type}/{year}/{month}/{day}/{natural identifier}/{domain}/{version}/{level}/{point in time}/{language}/{format}

A second example is the HTTP URI template used in Ireland.

COUNTRY/LEGAL INSTRUMENT	ELI TEMPLATE/EXAMPLE
Ireland	/eli/{year}/{type}/{Natural identifier}/{Level 1…}/{Point in Time}/{Version}/{language}
Motor Vehicle (Duties and Licences) Act 2012 (10/2012)	http://www.irishstatutebook.ie/eli/2012/act/10/enacted/en

In addition, examples of ELI HTTP URI templates used in France can be found here: http://www.eli.fr [1].

Finally, examples of ELI HTTP URI templates developed by the Publications Office can be found in the specific section on URI templates on the Metadata Registry [2] website.

[1] http://www.eli.fr/en/constructionURI.html
[2] http://publications.europa.eu/mdr/eli/documentation/uri_templates.html

Good practice 6:
Designing URI template(s) to stay as close as possible to existing citation practice

Problem statement

Legislation is cited in different ways depending on the country, type of legal instrument and user. For example, legislation may be cited according to: the title (short or long); the number of the legislation; the date of adoption or publication. Citation practices are used to identify legal information and therefore the design of URI templates should comply with existing citation practices. If publishers of legal information do not follow this approach they greatly reduce the potential to share, reuse, cite and link legal information.

Recommendation

When designing URI patterns, publishers should stay as close as possible to existing citation practices. This can be achieved by using any of the available elements of the ELI URI template in any order. It is important that URIs be flexible and user friendly. Elements found in ELI URI templates that are commonly used in citation practices are: year, type of legislation, identifier and version [1]. See the following French example.

Vue le décret n° 2014-1169 du 10 octobre 2014 modifiant diverses dispositions réglementaires du code de la défense

ELI HTTP URI: http://www.legifrance.gouv.fr/eli/decret/2014/10/10/DEFD1415169D/jo/texte

A relevant standard for ELI implementation is the specification of the URI described in the Internet Engineering Taskforce Request for Comment 3986, 'Uniform Resource Identifier (URI): generic syntax' [2]. Further guidance on how to effectively use URIs in the context of the semantic web is given by W3C in the document *Cool URIs for the semantic web* [3]. Additional principles that should be applied in designing URI templates are described in the document *10 rules for persistent URIs* [4].

BENEFITS

Access to legislation
Improved discovery of legislation

Interoperability
Greater interoperability by means of common URI design patterns

Quality and reliability
Ensures availability of quality URIs

DOs

- Design ELI URI template(s) close to citation practice
- Ensure that URIs are human readable/reader friendly

DON'Ts

- Add elements in the URI template that are not necessary to identify a piece of legislation in a unique way

(1) Examples of URI patterns can be found on the ELI website.
(2) https://tools.ietf.org/rfc/rfc3986.txt
(3) http://www.w3.org/TR/cooluris
(4) https://joinup.ec.europa.eu/community/semic/document/10-rules-persistent-uris

Good practice 7:
Modelling HTTP URIs by treating each piece of legislation as a unique resource

Problem statement

If one HTTP URI resolves to more than one piece of legislation, a request for that HTTP URI will receive in response all the pieces of legislation resolved by that URI. This is called the URI homonym problem and can lead to confusion and interoperability conflicts as it might not be clear what resource is associated with a URI.

Recommendation

When identifying legislation by means of HTTP URIs it is important to first ensure that each HTTP URI identifies each piece of legislation in a unique way, i.e. publishers should ensure that a given HTTP URI will identify a single piece of legislation. Difficulties may typically occur when:

- the legislation does not have a number;
- the same number is attributed to more than one piece of legislation.

If the legislation does not have a number, it is recommended that one be attributed in the URIs. This is the approach followed by Luxembourg, where legislation is cited without a number as the example shows:

Règlement grand-ducal du 2 février 2015 portant organisation de la Conférence nationale des élèves.

The URI identifier for this act includes 'n7' in order to uniquely identify it.

http://eli.legilux.public.lu/eli/etat/leg/rgd/2015/02/02/n7

If the same number is attributed to more than one piece of legislation, it is important that the URI is designed in such a way that it includes enough elements to distinguish the two pieces of legislation, for example by including dates of adoption, the author, jurisdiction, etc. The choice of which elements to select is left up to the publishers of legal information, since each legal system has its own specificities.

BENEFITS

Access to legislation
Unambiguous assignment of URIs to legislation resources makes the legislation more discoverable

Interoperability
Reduced risk of interoperability conflicts

DOs

- Design a mechanism to assign additional numbers or character strings to disambiguate URIs if they lead to the same texts
- Use aliases in order to strike a balance between consistency of URI scheme and reality

DON'Ts

- Try to model an official journal; treat legislation as resources by themselves

Good practice 8:
Taking into account the concepts of the ELI ontology: legal resource, legal expression and format

Problem statement

ELI describes legal information based on the well-established FRBR model and distinguishes between the concepts of:

- 'legal resource' or 'work' (intellectual or artistic creation);
- 'legal expression' (the intellectual or artistic realisation of a work, typically in a text, a sequence of signs); and
- 'format' or 'manifestation' (materialisation of one of the expressions in a file).

Each time a 'work' is realised it takes the form of an 'expression'. The physical embodiment of an expression is a 'format', or 'manifestation'.

In line with the commonly known best practices for linked data, such a structure enables the retrieval of various resource representations via content negotiation (cf. Good practice 12), but to do so it is necessary to ensure that the HTTP URIs that identify the legal information also follow this structure.

Recommendation

While the use of the FRBR structure should be transparent to the user when navigating through the URIs, publishers of legal information should follow the FRBR structure when designing the HTTP URIs.

Assigning URIs to abstract legal resources, independently of the temporal version of an act, provides the opportunity to group basic acts and consolidations together.

For example:

- eli/etat/leg/loi/2011/04/08/n2 — identifies the abstract basic act 'Loi 1979 04 16';
- eli/etat/leg/loi/2011/04/08/n2/jo — identifies the act as it was originally published;
- /eli/etat/leg/rgd/2011/04/08/n2/consolide/20120830 — identifies a consolidated version of this act.

BENEFITS

Access to legislation
Allows retrieval of various legal resources representations

Interoperability
Contributes to the creation of the 'web of data' since the FRBR hierarchy allows provision of the set of metadata at each level

DOs

- Take into account the ELI ontology structure when designing the ELI URIs

DON'Ts

- Structure ELI HTTP URIs following a representation other than the FRBR model

Good practice 9: Considering the right level of granularity when identifying legislation

Problem statement

Citations to legal information can include references to subparts of a legal act. However, the production workflows are generally designed to handle identification of an act as a whole. The identification of an act without identifying its subparts, while a key step in ELI implementation, can be a limitation for the reuse of legal information. For example, users might want to directly cite a specific provision of a piece of legislation when using social media or other online tools, without providing a reference to the legislation as a whole.

Recommendation

When implementing ELI it is important to identify the right level of granularity of the HTTP URIs. The ELI ontology published on the Metadata Registry ([1]) allows publishers of legal information to model the components of a legal act, at any level of precision (e.g. chapter, article, paragraph, etc.) by using the class 'eli:LegalResourceSubdivision' and its property 'is_part_of'.

For example, the United Kingdom has developed a URI model that considers the 'section' parameter. This parameter refers to particular sections, articles and regulations within a piece of legislation.

The template of the URI including this parameter is the following:

- http://www.legislation.gov.uk/id/{type}/{year}/{number}[/{section}]

The identifier URI for Act of the Scottish Parliament 2014, Number 2 is:
- http://www.legislation.gov.uk/id/asp/2014/2

The identifier URI for Section 3 of the same document is:
- www.legislation.gov.uk/id/asp/2014/2/section/3

In general, if the production workflow is based on XML structures it may be relatively easy to determine the more granular levels by processing the XML records. This may require, however, that unique identifiers be assigned to each XML element corresponding to a section, with impacts on the publishing workflow.

While the implementation of ELI for subparts can typically be done in the second phase of the project, it is important to integrate it into the design of HTTP URIs from the beginning if such implementation is envisaged.

([1]) http://publications.europa.eu/mdr/eli/documentation/ELI_Ontology-v1.0.htm

BENEFITS

Transparency
Improved understanding of the context in which legal information exists

Access to legislation
Increases the type of information that is searchable and indexable

Interoperability
Allows better interaction between information systems

Savings
Increased interoperability between IT systems can lead to reuse of information, and thus creates opportunities for cost optimisations

DOs

- Design ELI URIs that take into account subparts of legal acts
- Keep in mind the way external users will want to use the URI, for example in social networks or in value-added services

DON'Ts

- Assign ELIs only at the level of acts and ignore the benefits of more granular identification and description
- Use anchors in the HTML code

Good practice 10:
Following common good practices for HTTP URIs

Problem statement

Designing URI templates in isolation, without learning from the experiences of those who have already done it, increases the risk of repeating the same mistakes, duplicating analysis and wasting resources by solving problems that have already been addressed by others.

Recommendation

While it is not possible to provide a fit-for-all URI template, there are a number of good practices that publishers can follow in order to foster the access to and reuse of legal information. Key good practices include:

- avoiding using more elements than necessary to identify legal information in a unique way;
- staying as close as possible to existing citation practice (cf. Good practice 6);
- implementing the FRBR hierarchy in the URI scheme (cf. Good practice 8);
- avoiding using elements that tend to change in HTTP URIs.

These elements are important to guarantee the persistence of the URIs. Furthermore, when URIs are deprecated, meaning that they should no longer be used, they should nevertheless be maintained both in terms of the actual URI and the resource they identify. Once minted, a URI should not be deleted and a redirection mechanism should be put in place (cf. Good practice 15). Where URIs identify resources that are versioned, new URIs should be minted for each new version of the resource, along with a URI for the latest version.

DOs

- Use or create an integrated workflow for publication that includes the assignment of identifiers
- Use elements in the URI template(s) to indicate subparts of legislation (e.g. an article of a piece of legislation you want to identify)
- Create pleasing and recognisable identifiers so that people will understand what they can expect to find when they click on it

DON'Ts

- Use more elements than necessary in the URI template
- Include elements in the ELI that might be prone to change
- Use different URIs to identify the same piece of legislation

BENEFITS

Quality and reliability

High-quality URIs which meet users' needs

Good practice 11:
Testing URIs against real data

Problem statement

When designing URI templates, publishers of legal information may face a number of challenges, such as being able to correctly identify subparts of legislation, being able to uniquely identify legislation that does not have a unique identifier or being able to handle dates where more than one type of calendar year applies. These difficulties may not immediately be apparent during the design phase of ELI HTTP URIs and it may not be possible to foresee all possible scenarios.

Recommendation

To reduce the risk of developing URI design templates that do not take into account all possible scenarios, it is recommended that tests be performed as early as possible. The tests should reveal issues related to the correspondence between URIs and the way legislation is cited; identification of consolidated acts (when applicable); and resolution mechanisms. Tests should also include simulations of probable scenarios that can happen during the publication process, such as: 1. create a new legal act; 2. update the content of an act by altering its structure; 3. replace an act; 4. amend an act; 5. consolidate an act. The tests should be designed to match the context of each implementer. For example, the Publications Office had to find a way to assign unique and persistent URIs to legislation that did not have an identifier, such as corrigenda, which led to the following URI template for corrigenda:

http://data.europa.eu/eli/{typedoc}/{year}/{natural_number}/corrigendum/{pub_date}/oj where 'typedoc' is the resource type of the corrected act and 'pubdate' is the date of publication of the corrigendum in the Official Journal.

BENEFITS

Access to legislation
Ensures that all possible scenarios are taken into account

Interoperability
Ensures quality of URI templates, which in turn increases trust and potential for others to rely on them

Quality and reliability
Reduced risks of wrongly identifying legal information

DOs

- Choose diverse datasets for testing purposes
- For each tested dataset, compare the obtained ELI URIs to the way people cite legislation
- Test all the activities related to your URI management processes
- Test as many scenarios as possible and check results with users from the business side

DON'Ts

- Start large-scale implementation without testing your ELI URI template(s)
- Test only main scenarios related to management activities of your URIs

STEP 3: Configure web server for ELI URI resolution

Publishers of legal information should design HTTP URIs in such a way that they resolve to legislation. Resolution means that if an HTTP request is issued (either by a human using a web browser or by a machine/an application) to a web server, this returns a representation of the resource, either:

- the metadata of the legal resource (in either a human-readable or a machine-readable way);
- the legal resource document itself (in one of its available formats, e.g. HTML, PDF, XML).

The resolution mechanism refers to the service provided by a web server that handles URI requests of HTTP clients and attempts to obtain a representation of the resource that the URIs identify.

Legislation resources must be associated with content information so that resolution of HTTP URIs leads to a response with data.

ELI does not impose a specific solution. Depending on different scenarios, publishers of legal information may decide to return metadata and content or one of the two. For example, if a publisher of legal information does not have metadata, but only the full text of the legislation in PDF format, ELI does not oblige the publisher to implement metadata. In this case the publisher of legal information may decide to only return the actual legal resource.

It is recommended that tests be carried out to uncover and correct any potential issues.

It is important to always schedule production deployments and to publicly announce such events well in advance. It is also useful for the publishers of legal information to offer a short description of the changes that will occur after deployment. By making such information publicly available, consumers of legislation are aware of upcoming changes and can be prepared.

Good practice 12:
Using content negotiation and/or identifiers for language and format variants

Problem statement

If legislation — or the metadata describing legislation — is available in multiple languages and formats, publishers of legal information need to decide how to identify these different languages and formats.

Recommendation

To identify language and format variants, different persistent URIs can be used, possibly in combination with content negotiation. As described in Good practice 8, different language versions of legislation can be considered to be different 'expressions', whereas different formats can be considered to be different 'manifestations' of the same 'work' following the ELI ontology. Minting different URIs for these language and format variants of the same 'work' is good practice, whereby only meaningful language and format indicators should be used. For example, the use of technology-specific extensions like .asp, .aspx and .jsp in URIs should be avoided as this is prone to change; meaningful format indicators like HTML, RDF and PDF should be allowed. For example, the following URI could be used to distinguish an English-language variant in HTML format:

http://data.europa.eu/eli/dir/2013/37/oj/eng.html.

Additionally, it may be useful to apply content negotiation, although this is not explicitly required by the ELI Council conclusions. Content negotiation is the process that takes place between a web server and a web agent to select and provide the best representation for a given response when there are more representations available. Such a mechanism allows a request to be sent to a web server and the desired specific representation to be specified in the header of the request. For example, requesting the resource http://data.europa.eu/eli/dir/2013/37/oj, and providing the 'Accept' parameter in the request header with the value 'text/html', and the 'Accept-Language' parameter with the value 'en', would allow the server to choose the appropriate response, which in this example could be http://data.europa.eu/eli/dir/2013/37/oj/eng.html.

BENEFITS

Access to legislation

Better access to legislation by allowing consuming applications to request the format that is most appropriate in a particular situation

Interoperability

Increases the flexibility of adding or removing representation formats of resources while preserving persistence of URIs

DOs

- Include meaningful format indicators in ELI URIs such as .html, .rdf, etc.
- Consider using content negotiation

DON'Ts

- Use technology-specific extensions in the ELI URIs like .asp, .aspx, .jsp, etc.
- Include website-specific elements in the URIs, such as session cookies, etc.

Good practice 13:
Providing service-level guarantees

Problem statement

Potential consumers of legal information might be reluctant to rely on URIs if:

- they do not have enough guarantees that URIs are resolvable and persistent; and
- they do not have a reasonable request–response time and an acceptable level of data quality.

Recommendation

It is recommended that service-level guarantees be offered as part of the URI policy. The service-level guarantee should be a commitment to offer:

- resolvable URIs;
- long-term persistence of minted URIs;
- a reasonable request–response time;
- reasonable quality of legislation.

The service-level guarantee should also provide information about how issues encountered by the legislation consumers will be reported, who will address reported issues and how the resolution will be communicated back to those who reported them.

DOs

- Plan to provide service-level guarantees
- Plan to monitor performance of the resolution mechanism and the provision of service-level guarantees
- Ensure consumers of legislation can access service-level guarantees
- Review the service-level guarantees on a regular basis and adapt whenever necessary

DON'Ts

- Allow poor performance of the resolution mechanism, as this will reduce the interoperability potential of your solution
- Ignore issues reported by consumers of legislation

BENEFITS

Access to legislation
Increased consumption of published legislation

Interoperability
Improved trust and thus reuse potential

Quality and reliability
Improved reliability of legal information published

Good practice 14:
Monitoring website usage and links to legislation on the web through statistics

Problem statement

Monitoring the website traffic might not always be a common practice among publishers of legal information. As a consequence, they have a very limited insight about uses of legal information and any problems encountered by the users. Without having a good overview of the users' experience, publishers of legal information might miss opportunities to correct errors and ultimately improve the service according to the users' needs.

Recommendation

It is recommended that tools be implemented to regularly analyse the website traffic to optimise and improve access to published legislation. Key indicators that should be monitored include: 1. visits: the number of visits to the website during a given time period; 2. unique visitors: the number of different people who visited the website during a given time period; 3. visitor demographics: the profile of the website's audience such as the geographic location, language, gender, age, etc.; 4. bounce rate and time on site: whether visitors find what they are looking for in the website or if they leave it immediately; 5. type of sources: it segments the traffic by specific sources and mediums, such as search engines, referring sites, email or custom campaigns; 6. access by redirections of URLs vs access by redirections by a search engine: users that look for legislation according to the URI templates; 7. social media interactions: amount of visitors that interact with social media profiles.

BENEFITS

Quality and reliability

A careful watch of key traffic metrics will allow the provision of the stated service-level guarantees, and will constantly optimise and improve the quality of service

DOs

- Implement website traffic analytics tools
- Watch key performance indicators to constantly look for improvements

DON'Ts

- Ignore surveying the website traffic, as this can lead to a lack of persistence of services

Good practice 15:
Monitoring service levels and persistence of URIs

Problem statement

The added value that ELI brings in terms of increased persistence and common use of metadata to describe legal information could be greatly hampered by the absence of a reliable online service. In fact, if users are not able to access legal information in a consistent and timely fashion due to poor service performance, their level of trust in the service will decrease and they will tend not to rely on it in the future.

Recommendation

It is recommended that tools be deployed that allow the monitoring of website traffic. Such monitoring will allow the discovery of issues such as broken links or long request–response times. By finding such issues in time, the appropriate actions can be taken in a timely manner to remedy the situation, thus providing the guarantee of a quality service.

The analysis of weblog files can provide valuable insights into the web server's performance. GoAccess ([1]) is a popular choice for this task. It can run in a terminal window, and is useful for quickly analysing and viewing web server statistics in real time. It can be used to generate HTML, JSON and CSV reports from the weblogs.

Besides monitoring performance of the web traffic generated by visitors, it is recommended that load and stress tests be performed. Such tests allow understanding of the behaviour of a system under a specific expected load and its limits. Some of the most popular tools for load and stress testing are curl-loader ([2]), ApacheBench ([3]) and Httperf ([4]).

It is recommended that the response time to issues reported by the legislation be monitored and statistics gathered about the resolution of issues. This will allow the optimisation of the issue reporting–resolving mechanism.

DOs

- Monitor website service levels. Fix any observed or reported issue in a timely manner
- Review and optimise your issue-resolution mechanism

DON'Ts

- Ignore low request–response times and broken links
- Ignore issues reported by consumers of legislation

BENEFITS

Quality and reliability

Better quality of the service thanks to prompt replies to any occurring issues

([1]) http://goaccess.io
([2]) http://curl-loader.sourceforge.net
([3]) http://httpd.apache.org/docs/2.2/programs/ab.html
([4]) https://github.com/httperf/httperf

Pillar 2:
How to design an ELI metadata schema

STEP 1: Capture requirements for metadata

The second ELI pillar is described in the annex to the Council of the European Union's conclusions [1] which explains that:

'While a structured URI can already identify acts using a set of defined components, the attribution of additional metadata established in the framework of a shared syntax will set the basis to promote interchange and enhance interoperability between legal information systems. By identifying the metadata describing the essential characteristics of a resource, Member States will be able to reuse relevant information processed by others for their own needs, without having to put into place additional information systems. Therefore, while Member States are free to use their own metadata schema, they are encouraged to follow and use the ELI metadata standards with shared but extensible authority tables, which permit to meet specific requirements. The ELI metadata schema is intended to be used in combination with customised metadata schemas.'

Metadata provides the means for describing, classifying, linking, finding legal information as well as connecting it to other pieces of information and enabling users to access and reuse it. It includes, for example: title, date of adoption, date of signature, and author of a given piece of legislation. Having the ability to search and access legal information through metadata makes it much easier for someone to locate and use a specific piece of legal information. Moreover, connecting legal information with external sources (e.g. providing references to a controlled vocabulary [2]) to describe the subject, or a geographical location related to the legal information, or the publishing body) provides a wider context for the published legislation and makes it more usable and, in time, more used. In order to ensure that the metadata provided meets the users' needs, publishers of legal information should gather user requirements using the techniques highlighted in Pillar 1, Step 1 and, if necessary, update the business case (cf. 'Organisation and policy', Step 1).

Relevant aspects that publishers of legal information should investigate are:

- **the reasons** users access legal information, such as personal or professional reasons;
- **the way(s) users access** legal information, for example whether they type keywords into general search engines or reference numbers directly on legal information websites;
- **what information users are looking** for and what obstacles they encounter, for example with regard to the identification of relationships between various pieces of legislation or consultation of specific articles;

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(1) OJ C 325, 26.10.2012.

(2) http://publications.europa.eu/mdr/eurovoc/index.html

- **profiles of users**, whether they belong to the private or public sector, whether they are mainly national, international or internal users (user studies, website statistics on usage and links to your legislation on the web);
- **uses of legal information**, for example whether users cite a piece of legislation, establish cross-references between different types of data, reference legal information via social media or develop mobile applications;
- **expectations** users have in terms of the availability and quality of legal information;
- **types of legal resources** that are of interest for end users, including legacy contents;
- **external information resources** to which the published content can be connected.

Thanks to the requirements collected, publishers of legal information will be able to identify what legal information should be described according to ELI-compliant metadata. For example, if users are mainly interested in consulting in-force legislation, an option could be to focus on applying ELI-compliant metadata to this collection first. Another option could be to apply ELI-compliant metadata not only to legislation, but also to relevant information that helps users to understand what the legislation is about, such as summaries of legislation.

Good practice 16:
Following a user-centered design taking into account type of users

Problem statement

Publishers of legal information could implement the ELI second pillar without knowing who will use the service and for what purposes. Discarding such information increases the risk of implementing a solution that does not answer the needs of those who will consume legal information, therefore reducing its potential for use.

Recommendation

It is recommended that the metadata be defined, taking into account both user profiles (such as legal professionals, journalists, academics) and their requirements, by looking at: 1. what metadata helps users find what they are looking for; 2. for what purposes users want to consult legal information; 3. what formats can help consumers of legal information to use it further; 4. how often legal information is consulted; 5. what information users need to understand legal information.

Luxembourg, for example, gathered a list of queries that users would like to make, such as: all acts that implement a directive (or an article thereof) on a specific date, after deadline or partially; all acts adopted by a ministry in a given year, with a specific subject (such as 'water'); acts based on parliamentary number or date of signature; all acts adopted that are based on a particular article of the constitution.

Collecting user requirements greatly helps publishers of legal information to better understand what ELI metadata should be implemented. For example, if users look for legal information by the legislation number, it is a good idea to include the metadata 'eli:number'. Similarly, if users are interested in the date of entry into force, the metadata 'eli:date_entry_into_force' can help. To then ensure that the metadata actually meets the identified requirements, publishers of legal information should monitor website usage based on predefined metrics (cf. Good practices 14 and 15).

DOs

- Carry out an in-depth investigation of user requirements
- Document user requirements in the business case
- Identify key metrics to measure whether the needs are met

DON'Ts

- Engage in development without performing an appropriate analysis of user requirements

BENEFITS

Quality and reliability
Provides services that meet user requirements

STEP 2: Design and publish controlled vocabularies and metadata schemas

Designing a metadata schema in line with ELI specifications is a fundamental step to enable publishers of legal information to '... identify and exchange legal information originating from regional and national authorities at the European level', as explained by the ELI Council conclusions ([1]).

The design of the metadata schema should begin with the end in mind: prepare the publication of ELI metadata in the HTML pages of the publisher's website. To do so, the following steps are necessary.

1. Select from the ELI ontology ([2]) the properties that will be published in the HTML page. The choice depends mainly on the availability of the metadata in the publisher's information system. The set of metadata properties may vary depending on the document type. For metadata that uses a controlled vocabulary, decide on the controlled vocabulary to use. The selection of metadata, the choice of controlled vocabularies, and the naming convention for ELI URIs decided for Pillar 1 form the 'ELI application profile' of the publisher for the purpose of ELI metadata publication in HTML pages.
2. Create the controlled vocabularies that do not already exist, typically the list of document types. The ELI annotation tool ([3]) contains a converter to create the correct file format from Excel spreadsheets.
3. Make the ELI application profile and controlled vocabularies available. A change-and-release management procedure should be set up to ensure the quality of the publication. Definitions of the ELI application profile classes and properties should be translated to ensure they are understood by the citizens in the publisher's own language. Controlled vocabularies should be made available in HTML, CSV and RDF/SKOS format to ensure their reusability.
4. Map the data model already managed by the publisher to the ELI application profile. This prepares the data conversion of the existing metadata to the ELI application profile and the publication of ELI metadata. Metadata that is already managed by legal publishers may be stored in a content management system, in a specific application or in the XML of documents. The publishers should map:
    a) existing metadata fields with the corresponding fields in the ELI application profile to describe the equivalence between the two data models;

([1]) OJ C 441, 22.12.2017, pp. 8-12.

([2]) http://publications.europa.eu/mdr/eli/documentation/ELI_Ontology-v1.1.htm

([3]) http://eur-lex.europa.eu/eli-register/resources.html

b) values of the metadata in the existing information system with their value in the controlled vocabularies selected or created for the ELI publication.

Publishers who already have a metadata schema can implement ELI to enrich their internal metadata schema and to publish more ELI metadata. These publishers will first modify their annotation process for the new legal resources and eventually decide on a process to enrich the metadata of their legacy of legal resources.

For publishers who do not have a metadata schema, or have only a very limited one, the implementation of ELI can be an opportunity, as the ELI ontology is a state-of-the-art ontology for legal resources. It builds on the know-how of European official journals, and is also compliant with the most recent bibliographical metadata schemas used by libraries for open data catalogue publications (FRBR). Publishers can select the properties of the ELI ontology they want to use, but can also enrich the ELI metadata model to fulfil their specific needs. The resulting metadata schema will be used for the internal management of legal metadata.

Further information on the metadata specifications can be found in the Documentation ([1]) section on the Metadata Registry ([2]) website.

([1]) http://publications.europa.eu/mdr/eli/documentation/index.html
([2]) http://publications.europa.eu/mdr/eli/index.html

Good practice 17:
Considering the right level of granularity when creating metadata

Problem statement
ELI metadata can describe legal acts at different levels of granularity, and publishers of legal information can choose to implement the metadata they want. For example, it is possible to identify legal information at document level using 'eli:type_document' or define a more granular description. The first approach is easier to follow, because it requires less metadata to be generated. However, generating more granular metadata usually meets user requirements better.

Recommendation
It is recommended that the appropriate level of granularity, in line with user requirements and needs, be chosen when creating metadata that describes legal information. More granular metadata offers a greater variety of information that can be reused, enabling greater levels of technical manipulation. Legal considerations can also come into play: legal publishers should refrain from disseminating metadata for which they are not totally sure they have an accurate value. For example, knowing whether or not an act is 'in force' is not trivial. It is important to strike the right balance between a level of granularity that allows publishers of legal information to organise and maintain content in the most appropriate way and one that helps consumers of legal information to find and reuse what they are looking for.

BENEFITS

Access to legislation
More functionalities available to access legislation

Interoperability
Further exchange of information allowed

Quality and reliability
More detailed information available

Development of new services
More information is provided and more functionalities can be built on top of it

DOs

- Create sufficient metadata at various levels of granularity
- Create sustainable metadata for future purposes balancing the cost of creation and maintenance

DON'Ts

- Create metadata with insufficient details
- Disseminate inaccurate metadata

Good practice 18:
Structuring information by linking it to other documents

Problem statement

Legislation does not exist in isolation. For example, there is national legislation that transposes EU directives or national legislation that amends or repeals other national legislation. The ability to establish the relationship between different pieces of information, such as establishing the link between basic acts and related amendments, is key for supporting a good understanding of the evolution of a given law. If publishers of legal information do not take the necessary steps to ensure that legal information is linked to relevant data, it is more difficult for users to understand, share and reuse legal information.

Recommendation

Publishers of legal information should design the ELI metadata schema making explicit relationships between acts. Examples of relationships that can be expressed using the ELI metadata schema include citations, amendments, consolidations, commencement and transposition of EU law into national laws. ELI metadata helps publishers of legal information to make explicit a variety of relationships between legal resources, boosting the analysis that can be carried out on legal information. Examples of available ELI metadata include:

- 'related to', a generic property defined as a link to 'a somehow related other document, not necessarily a legal resource';
- 'consolidates', defined as a link between a consolidated resource and the amendments it incorporates;
- 'transposes', defined as a link between a piece of national legislation and a transposed EU directive;
- 'changes', to indicate legal modifications between two acts, along with more precise properties such as 'repeals', 'amends' or 'commences'.

Linking information to other documents can also help in developing innovative applications, for example to create visualisations that enable a quick overview of relationships between documents.

BENEFITS

Access to legislation
Easier identification of the wider context to which legal information relates

Interoperability
Improved links and exchange of information

DOs

- Reuse vocabularies whenever possible, as identified in the ELI metadata specifications

DON'Ts

- Miss the opportunity to make the links between legal resources explicit

STEP 3: Annotate legal resources (automatically or manually)

In this step, publishers of legal information should consider one of the following two use cases.

1. Metadata is available in the information systems of the publisher.
2. No metadata, or poor metadata, is available and the publisher needs to enrich the description of legal resources to implement ELI.

In the first case, the publisher will have to carry out data conversion from the existing metadata to the ELI metadata. To do so the publisher should make sure the following elements are available.

- Mappings between existing metadata schemas and the ELI ontology elements used as part of the ELI application profile.
- Mappings between existing metadata values and values of the controlled vocabularies used as part of the ELI application profile.
- Data conversion rules that might be needed.

If all the above is available, the work of the publisher will be to prepare the scripts which will execute the data conversion. The output of the scripts will be used to add RDFa tags in web pages. The XML serialisation of ELI ([1]) and the associated conversion stylesheet can help at this stage.

In the second case, the publisher will have to create metadata. Metadata can be created manually (the choice of the large majority of ELI implementers in this situation) or (semi-)automatically. The ELI annotation tool ([2]) allows ELI notices of legal resources to be manually created.

The publisher can add metadata during the editorial process of legislation. At the end of the editorial process a last step should allow verification and enrichment of this metadata. Legal resources will be annotated with this metadata. The ELI validator ([3]) can help in validating the final ELI metadata inserted in the HTML pages.

The automatic process of creating metadata can be used for some of the metadata, and most often to enrich the metadata of the legacy contents. In such cases, manual work is necessary to detect and correct any errors generated during the metadata extraction process.

([1]) http://publications.europa.eu/mdr/resource/eli/eli_xml/
([2]) http://eur-lex.europa.eu/eli-register/resources.html
([3]) http://publications.europa.eu/eli-validator/

Good practice 19:
Creating metadata as part of the workflow, including validation services

Problem statement

When metadata that describes legal information is created manually after legislation is published online, publishers of legal information run the risk of causing errors and incoherencies because manually created metadata tends to be less accurate, especially in the absence of validation services.

Recommendation

Now that legal information is increasingly available in digital form, publishers of legal information have the opportunity to apply metadata at the time of creation as part of the workflow and in a semi-automated way. This approach ensures that metadata can be used throughout the publishing process, enabling publishers of legal information to concentrate human intervention where it matters. The machine-generated metadata will typically include information that can be gleaned from the document structure or process stage (e.g. title, identifiers, dates, author). The human involvement can then focus on adding information that it is not possible to extract automatically from the document (e.g. subject classifications, links to other legal information) and on carrying out quality controls. Validation can also be incorporated into the automated tools, including spell checks, numbering, consistency of numbering and link checking.

To support the quality control of the publishing process, an ELI metadata validation tool is available ([1]) to validate the compliance of RDFa annotations with the ELI ontology.

DOs

- Deploy workflow tools that capture metadata along the way
- Automate validation as much as possible
- Ensure that content specialists focus on carrying out intellectual analysis and quality assurance
- Rely on ELI tools: ELI/XML, ELI validator, ELI annotation tool

DON'Ts

- Keep metadata for internal use only
- Generate metadata at the end of the publishing process
- Attribute all metadata in a manual way

BENEFITS

Quality and reliability
Increased reliability of the workflow by minimising human intervention to a few selected tasks

Development of new services
Ability to reuse the process for different information workflows

Savings
Balanced approach between automatic and manual intervention when attributing metadata, which leads to: effectiveness; decreased error rate; increased reliability of the process as a whole

([1]) http://publications.europa.eu/eli-validator/

Good practice 20:
Generating standard metadata formats

Problem statement

Metadata is key to ensuring that legal information can be accessed and reused across information systems. Describing legal information with metadata allows it to be understood by both humans and machines in ways that promote interoperability. However, such potential for interoperability may be hampered in the absence of standard metadata, because only a few will know the meaning of the customised metadata and others will find it difficult to understand and process it.

Recommendation

It is recommended that ELI-compliant metadata schemas be used to ensure greater accessibility and interoperability across legal information systems as well as greater effectiveness of the publication process.

The creation of formal metadata through adherence to ELI metadata is essential for managing and sharing legal information. ELI metadata enables interoperability among legal information systems that use the same metadata standards, greatly enhancing the exchange of legal information. To ensure interoperability, publishers of legal information should use consistent ontologies throughout the legal information and metadata should use the same terms for describing legal information.

The ELI metadata can be accessed via the Publications Office's Metadata Registry website: http://publications.europa.eu/mdr/eli.

BENEFITS

Access to legislation
Facilitates search criteria for search engines

Interoperability
Allows the exchange of information in a standard way

Development of new services
Facilitates the development of added-value services

Savings
Increased degree of automation for querying, accessing and exchanging information

DOs

- Use ELI-compliant metadata scheme to describe legal information
- Make available metadata for reuse

DON'Ts

- Use custom-made metadata while the same is available in ELI

Pillar 3:
How to render ELI metadata machine-reusable

STEP 1: Add RDFa tags to web pages to facilitate parsing and extraction

If publishers of legal information already publish legal information on web pages using HTML or create new web pages, there is a preliminary step that they are recommended to take before adding machine-readable markups.

This step entails ensuring that the web pages are compliant with standard syntax conventions so as to facilitate metadata parsing and extraction. In fact, if publishers of legal information add microdata or RDFa without ensuring that web pages are compliant with standard syntax convention, they make it more difficult for consumers of legal information to extract and parse the metadata.

The standard syntax conventions that publishers of legal information are invited to use for their web pages are XHTML (eXtensible HyperText Markup Language) ([1]) or HTML5 ([2]).

Once the web pages are compliant with these standard syntaxes, the next step entails serialising the metadata so as to make it machine readable, annotating the HTML pages with RDFa ([3]) markup.

An ELI annotation tool is available ([4]) to manually create RDFa-enriched HTML pages after filling in a form with fields corresponding to the ELI ontology.

An ELI RDFa validation tool is available ([5]) to help publishers of legal information ensure the validity and correctness of the published legislation.

([1]) http://www.w3.org/TR/xhtml1
([2]) http://www.w3.org/TR/html5
([3]) http://www.w3.org/TR/
([4]) http://eur-lex.europa.eu/eli-register/resources.html
([5]) http://publications.europa.eu/eli-validator/

Good practice 21: Annotating legal information with RDFa markup

Problem statement

When selecting metadata formats to describe legal information, publishers of legal information have many options to choose from. How publishers choose to represent metadata is a primary factor in someone else's ability to use it in the future. Thus it is important to think carefully about what format will be best for managing, sharing and preserving the metadata.

Recommendation

In order to promote the reuse of metadata, publishers of legal information should make it available in a non-proprietary format. To enable further use, legal information should be annotated with RDFa markups.

Here are some examples of how metadata published in RDFa format can make legal information more discoverable and usable.

- Metadata (e.g. title, author, publication date, description, type of document) can be read as structured data and transferred, categorised, sorted, etc.
- Metadata can be read and harvested by tools.
- The meaning of the harvested metadata is preserved.

In order to make legal information reusable, metadata should be accurate and up to date. Publishers of legal information should ensure that legal information is annotated with quality metadata.

Example: In the context of transposition of EU legislation, ELI metadata of legal resources published in RDFa by national public administrations can be automatically harvested by the Commission, which makes the transposition notification mechanism more efficient. This is a concrete example of the smart use of technology in line with the objectives of the better regulation guidelines, since it helps the Commission monitor the efforts of the Member States to ensure their legislation is in line with EU law.

BENEFITS

Transparency
More legal information available in a standard way

Interoperability
Improved access and exchange of information through standard formats

Savings
Increased degree of automation for querying, accessing and exchanging information

DOs

- Use RDFa as the open machine-readable standard format

DON'Ts

- Use proprietary formats

Good practice 22:
Dealing with legacy content

Problem statement

Legacy content is often only available in the form of static files such as Microsoft Word documents or PDFs, which makes it a challenge to process and publish metadata that conforms to ELI. Publishers of legal information have to identify a minimum set of ELI ontology elements to describe the legacy content and find ways to process it, transform it into structured content, annotate it and publish it.

Recommendation

It is recommended that a set of elements be identified in the ELI ontology that can be extracted from the legacy content. For example, the elements relating to the type of document, date of signature, date of publication and language can be used to design ELI URI patterns. Further elements of the ELI ontology that could enrich legacy content are the title, the subject of the document, description, etc.

The processing of legacy content depends on several factors, such as what metadata can be extracted and processed, language constraints, and the cost and availability of manual work. The publishers of legal information can choose to process legacy content manually or to automate the work to extract metadata values from legacy content. Manual data cleansing should be provided to increase the quality of the metadata extracted automatically. The ELI annotation tool [1] can help in the manual metadata-creation process.

BENEFITS

- Improved data quality, by ensuring that legislation is made available in a structured way
- Increased accessibility, by making legacy content available in a structured way
- Increased interoperability, by making the legacy content available in a machine-readable format that respects the ELI ontology

DOs

- Enrich legacy content with a set of metadata that can be used by local information systems as well as to publish ELI metadata
- Make legacy content available in a machine-readable format

DON'Ts

- Publish legacy content without descriptive metadata

[1] http://eur-lex.europa.eu/eli-register/resources.html

PART 3:

Beyond the ELI pillars

ELI is one approach to enhancing the discoverability and reuse of legal information. It is an easy first step into publishing legislative metadata on the web of data. Other actions can, however, be taken beyond ELI to make data more easily reusable, or to consider ELI as a starting point to improve the internal publishing workflow.

Align metadata values with other vocabularies (Eurovoc)

Alignment of local controlled vocabularies (e.g. local thematic controlled vocabulary, types of documents, etc.) as part of an ELI application profile to EU controlled vocabularies (e.g. Eurovoc ([1])) will increase the potential for interoperability between publishers of legislation.

In particular, alignment of legal subjects on a common controlled vocabulary will allow a subject-based cross-national search of legislation.

Tools such as the ones listed below provide the necessary functionalities to create and publish mappings between controlled vocabularies.

- SILK: a tutorial on the use of SILK to align controlled vocabularies is available on Joinup ([2]).
- Semantic Interoperability Community ([3]) and mapping.semic.eu.
- OnaGUI — Ontology Alignment Graphical User Interface ([4]).

Use other metadata dissemination channels

ELI mandates the use of RDFa as dissemination syntax. RDFa is a good choice from a publisher's point of view because it relies on the existing web portal infrastructure and is relatively easy to implement. However, this publishing technique requires data consumers to crawl and parse all HTML pages to get the full set of ELI metadata from a given portal. Other dissemination techniques can be used to facilitate the reuse of metadata.

- Content negotiation through the ELI URI, returning RDF files, while not allowing the data to be retrieved as a complete set, relieves the data consumers from parsing RDFa markup.
- Releasing RDF bulk files containing the full set — or large subsets — of ELI metadata allows data consumers to instantly get the full metadata content. However, this does not provide the most recent updates to the metadata.

([1]) http://eurovoc.europa.eu/drupal/
([2]) https://joinup.ec.europa.eu/document/tutorial-use-silk-aligning-controlled-vocabularies
([3]) http://semic.eu
([4]) https://github.com/

- Publishing the data using an API with proper documentation allows data consumers to retrieve the subset of the data they are interested in. This is usually a very large project, requiring a lot of resources and with other impacts on the information system. Publishers need to do the following.
  - Create a web application capable of rendering machine-readable metadata for legislation that should conform to the ELI-compliant metadata schema. In particular, they should make sure that the data is not locked into a proprietary API data model, but follows the principles of linked data and the ELI data model. The JSON-LD syntax ([1]) provides a good way to create API based on JSON and still linked-data compatible.
  - Publish API documentation. An example of such documentation is the one provided by the Publications Office, which can be found at http://data.europa.eu/eli.
- Opening the data in a SPARQL endpoint ([2]) is an alternative to creating an API. This is closer to the original ELI metadata work since it does not require that a new data model or the inputs and outputs of the API be specified.

Use ELI as an opportunity to improve publishing workflows

ELI focuses on the metadata description and the dissemination of legislation. Experience has shown that, while an ELI implementation project is progressing, it raises fundamental questions on the way the legislation is published.

- Why is metadata unavailable? How could they be made available?
- Why is it hard to have representations of the documents (PDF, HTML, XML)? Can the internal format of legislative document be enhanced?
- What are the impacts of ELI in terms of user navigation on the website?
- Could additional search facets or search criteria be proposed to search legislation on the website?

The use of Akoma Ntoso as an XML representation format for the documents can help streamline the workflow. Such an XML schema can store metadata in the document header, and the ELI/XML schema ([3]) can be integrated in an Akoma Ntoso proprietary metadata field.

The improvement of metadata quality, or the addition of new metadata fields, can be 'side projects' next to ELI implementation in order to

([1]) https://www.w3.org/TR/json-ld/
([2]) https://www.w3.org/TR/sparql11-protocol/
([3]) http://publications.europa.eu/mdr/resource/eli/eli_xml/

improve the overall quality of disseminated metadata. In particular, the legal subject ('eli:is_about') is an important item of metadata to enable a future subject-based cross-national legislation search. The creation of metadata can also be pushed early in the creation workflow, with the legislation subjects typically being set as soon as the draft legislation is written.

Working on metadata can also be an opportunity to consider metadata as first-class citizens in the publishing workflow, typically by storing and managing them in a central, dedicated database. This is the dissemination architecture used by the Cellar at the Publications Office, or by Casemates in Luxembourg. The central database stores metadata and document representations and is regarded as the master storage point on which other business applications can rely to search legislation, to search values in controlled lists or to display a document.

Make your legislation more visible for web search engines

Major search engines consume structured data from the web. However, they cannot read any structured metadata. They understand only information structured according to the schema.org ([1]) vocabulary. The schema.org vocabulary, until recently, did not contain any specific classes or properties to describe legislation.

ELI's goal is to improve legislation discovery and interoperability, not only at European scale for technology-aware data consumers, but also at web scale for the average internet user. This is why the ELI Taskforce has proposed an extension to the schema.org ([2]) vocabulary for web search engines to be able to read structured metadata about legislation. This extension is derived from the ELI ontology and includes in schema.org the ELI properties that are not already included in the core vocabulary.

The long-term typical use case of this extension is that a search engine would be capable of showing on a search-result page whether or not a piece of legislation is in force, and/or give the choice to a user to read the latest consolidated version or the original version.

To improve the discoverability of your legislation HTML pages you might want to include in them structured data markup using the schema.org vocabulary. Note that this means that the HTML pages would contain both ELI structured metadata and schema.org metadata. However, this gives no guarantee that the search engines will actually read the metadata, or do anything useful with it.

([1]) http://schema.org
([2]) http://schema.org/Legislation

ANNEXES

ANNEX I: Glossary

TERM	EXPLANATION
Act	A law enacted by a parliament or similar legislative body.
Akoma Ntoso	Akoma Ntoso is an initiative managed by the United Nations Department of Economic and Social Affairs (Undesa), Division for Public Administration and Development Management (DPADM) and the University of Bologna. Akoma Ntoso defines a 'machine-readable' set of simple technology-neutral electronic representations (in XML format) of parliamentary, legislative and judiciary documents. The objective is to provide a framework for the exchange of 'machine-readable' parliamentary, legislative and judiciary documents. Akoma Ntoso XML schemas make 'visible' the structure and semantic components of relevant digital documents so as to support the creation of high-value information services to deliver the power of ICTs to increase efficiency and accountability in the parliamentary, legislative and judiciary contexts. (Source: http://www.akomantoso.org)
Amendment	An effect that changes the text of legislation.
Availability	Availability is the extent to which the data can be accessed. This also includes the long-term persistence of data. (Source: https://joinup.ec.europa.eu/community/ods/description)
Chapter	A numbered level of division within an act or other legislation.
'Coming into force' date	The date on which a legislative provision or an effect comes into force.
Consistency	Consistency is the extent to which the data does not contain contradictions that would make its use difficult or impossible. (Source: Open Data Support)
Consolidation	Consolidation consists of the integration in a legal act of its successive amendments and corrigenda. (Source: http://eur-lex.europa.eu/content/help/faq/intro.html#top)
Content negotiation	Content negotiation is a mechanism of the HTTP protocol by which different documents (or 'representations') can be returned for the same URI. This is typically used to return a page in the language of the client for the same URL, or, in the context of the web of data, to return either a human-readable page or a machine-readable data file about the same URI.
Data integration	Almost any interesting use of data will combine data from different sources. To do this it is necessary to ensure that the different datasets are compatible: they must use the same names for the same objects, the same units or co-ordinates, etc. If the data quality is good this process of data integration may be straightforward but if not it is likely to be arduous. A key aim of linked data is to make data integration fully or nearly fully automatic. Non-open data is a barrier to data integration, as obtaining the data and establishing the necessary permission to use it is time-consuming and must be done afresh for each dataset. (Source: http://opendatahandbook.org/glossary/en/terms/data-integration)

Database	(i) Any organised collection of data may be considered a database. In this sense the word is synonymous with dataset. (ii) A software system for processing and managing data, including features to extend or update, transform and query the data. Examples are the open source PostgreSQL, and the proprietary Microsoft Access. (Source: http://opendatahandbook.org/glossary/en/terms/database)
Dataset	Any organised collection of data. 'Dataset' is a flexible term and may refer to an entire database, a spreadsheet or other data file, or a related collection of data resources. (Source: http://opendatahandbook.org/glossary/en/terms/dataset)
Dereferencing	The act of retrieving a representation of a resource identified by a URI. (Source: http://www.w3.org/)
Discoverability	It is not enough for open data to be published if potential users cannot find it, or even do not know that it exists. Rather than simply publishing data haphazardly on websites, governments and other large data publishers can help make their datasets discoverable by indexing them in catalogues or data portals. (Source: http://opendatahandbook.org/glossary/en/terms/discoverable)
Division	A term we use to denote any one of the hierarchical levels into which a piece of legislation may be divided.
Dublin Core	The Dublin Core Metadata Initiative, or 'DCMI', is an open organisation supporting innovation in metadata design and best practices across the metadata ecology. DCMI's activities include work on architecture and modelling, discussions and collaborative work in DCMI communities and DCMI task groups, global conferences, meetings and workshops, and educational efforts to promote widespread acceptance of metadata standards and best practices. DCMI maintains a number of formal and informal liaisons and relationships with standards bodies and other metadata organisations. Dublin Core is the common name for a generic set of metadata elements that were originally conceived as a simple mechanism to describe physical or digital resources, but developed into one of the main metadata standards for the semantic web. (Source: http://dublincore.org)
Effect	Any impact that one legislative provision may have on another. The most familiar type of effect is an amendment that changes the text of the affected legislation, but there are also types of effect that do not change the text, such as where a provision is said to be 'modified' or 'applied'.
Extent	The term 'extent' when used in legislation refers to the jurisdiction(s) for which it is law.
Five stars of open data	A rating system for open data proposed by Tim Berners-Lee, founder of the World Wide Web. To score the maximum five stars, data must (1) be available on the web under an open licence, (2) be in the form of structured data, (3) be in a non-proprietary file format, (4) use URIs as its identifiers (see also RDF), (5) include links to other data sources (see linked data). To score three stars, it must satisfy all of (1)-(3), etc. The open definition requires data to score three stars in order to qualify as open, not requiring RDF or linking. This permits data of a wider variety of types and sources to be open, without the work of creating linking information. (Source: http://opendatahandbook.org/glossary/en/terms/five-stars-of-open-data)

Hierarchy	'Hierarchy' and 'hierarchical structure' are terms we use to denote the levels of division within a piece of legislation and the relationship between them. For example, the level of a cross-heading in an act comes below the part level in the hierarchy, but above the section level.
Human-readable	Data in a format that can be conveniently read by a human. Some human-readable formats, such as PDF, are not machine-readable as they are not structured data, i.e. the representation of the data on disk does not represent the actual relationships present in the data. (Source: http://opendatahandbook.org/glossary/en/terms/human-readable)
Identifier	The name of an object or concept in a database. An identifier may be the object's actual name (e.g. 'London' or 'W1 1AA', a London postcode) or a word describing the concept ('population') or an arbitrary identifier such as 'XY123' that makes sense only in the context of the particular dataset. Careful choice of identifiers using relevant standards can facilitate data integration. (Source: http://opendatahandbook.org/glossary/en/terms/identifier)
Interoperability	Interoperability, within the context of European public service delivery, is the ability of disparate and diverse organisations to interact towards mutually beneficial and agreed common goals, involving the sharing of information and knowledge between the organisations, through the business processes they support, by means of the exchange of data between their respective ICT systems. (Source: http://ec.europa.eu/isa/documents/eif_brochure_2011.pdf)
Legislation	The generic term for laws of any type. The terms 'piece of legislation' and 'item of legislation' are used within this guide to mean a whole legislative document of any type, for example an act or statutory instrument.
Linked data	Linked data is a set of design principles for sharing machine-readable data on the web for use by public administrations, businesses and citizens. (Source: Open Data Support)
Long title	Acts and measures have two titles, the 'short title' and the 'long title'. The 'long title' sets out the purposes of the act, sometimes at great length, whereas the 'short title' is a more convenient short form by which the act will usually be known. For example, the Petroleum Act 1998 (short title) has a long title that reads: 'An Act to consolidate certain enactments about petroleum, offshore installations and submarine pipelines.'
Metadata	Metadata is structured information that describes, explains, locates or otherwise makes it easier to retrieve, use or manage an information resource. Metadata is often called 'data about data'. (Sources: National information, standards organisation)
Machine-readable	Data in a data format that can be automatically read and processed by a computer, such as CSV, JSON, XML, etc. Machine-readable data must be structured data. Compare human-readable. (Source: http://opendatahandbook.org/glossary/en/terms/machine-readable)

OCR	Optical character recognition (OCR) is the mechanical or electronic conversion of images of typewritten or printed text into machine-encoded text. It is widely used as a form of data entry from printed paper data records, whether passport documents, invoices, bank statements, computerised receipts, business cards, mail, printouts of static data or any suitable documentation. It is a common method of digitising printed texts so that it can be electronically edited, searched, stored more compactly, displayed online and used in machine processes such as machine translation, text-to-speech, key data and text mining.
Open data	Open data is data that can be freely used, reused and redistributed by anyone — subject only, at most, to the requirement to attribute and share alike. (Source: Opendefinition.org)
Open format	A file format whose structure is set out in agreed standards, overseen and published by a non-commercial expert body. A file in an open format enjoys the guarantee that it can be correctly read by a range of different software programs or used to pass information between them. Compare proprietary. (Source: http://opendatahandbook.org/glossary/en/terms/open-format)
Original 'Basic act' or 'As enacted' or 'As made'	The original version of the legislation as it stood when it was initially adopted. No changes have been applied to the text.
Paragraph	A provision, usually numbered, constituting the lowest level of division in a schedule. (But note that the term 'paragraph' may also be used in legislation to denote certain levels of subdivision within a provision.)
Part	A division of the main body or a schedule in an item of legislation, usually forming part of a numbered sequence of parts. A part may be further subdivided hierarchically into chapters, cross-headings and numbered sections (or paragraphs, if in a schedule).
Preamble	Words appearing near the beginning of an act after the long title, stating the reasons for passing the act. The use of preambles is optional and they are now rare. Any preamble would appear in the introductory text.
Primary legislation	General term used to describe the main laws passed by the legislative bodies. It is to be distinguished from secondary legislation.
Processability	The processability of data is the extent to which it can be understood and handled by automated processes. (Source: Open Data Support)
Proprietary	(i) Proprietary software is owned by a company which restricts the ways in which it can be used. Users normally need to pay to use the software, cannot read or modify the source code, and cannot copy the software or resell it as part of their own product. Common examples include Microsoft Excel and Adobe Acrobat. Non-proprietary software is usually open source. (ii) A proprietary file format is one that a company owns and controls. Data in this format may need proprietary software to be read reliably. Unlike an open format, the description of the format may be confidential or unpublished, and can be changed by the company at any time. Proprietary software usually reads and saves data in its own proprietary format. For example, different versions of Microsoft Excel use the proprietary XLS and XLSX formats. (Source: http://opendatahandbook.org/glossary/en/terms/proprietary)

Provision	The term 'provision' is used to describe a definable element in a piece of legislation that has legislative effect. Most commonly in the help documentation and messages on this site it will be used to refer to a section (or corresponding element such as a paragraph in a schedule or an article in an order) but it can also refer to higher-level divisions such as parts or chapters.
Public sector information	Information collected or controlled by the public sector. (Source: http://opendatahandbook.org/glossary/en/terms/public-sector-information)
RDF	Resource description framework (RDF) is a model and a set of syntaxes for sharing data in the web. The basic concepts of RDF are described in the W3C recommendation 'RDF 1.1 Concepts and Abstract Syntax'. (Source: http://www.w3.org/TR/rdf11-concepts) A primer, aimed to convey basic knowledge required to effectively use RDF, is published by W3C under the title 'RDF 1.1 Primer'. (Source: http://www.w3.org/TR/rdf11-primer) Specific syntaxes for the expression of RDF are given in the documents 'RDF 1.1 XML Syntax' and 'RDF 1.1 Turtle', both published by W3C. While RDF is increasingly being used for the description of resources on the web, there are no easy ways to check validity of RDF data against specified patterns, the way that XML can be validated not just for valid syntax but also for conformity to application-specific rules. The W3C Working Group 'RDF Data Shapes' has been established with the mission 'to produce a W3C Recommendation for describing structural constraints and validate RDF instance data against those'. (Source: http://www.w3.org/2014/data-shapes)
RDFa	Resource description framework in attributes (RDFa) is a way of expressing RDF-style relationships using simple attributes in existing markup languages such as HTML. RDFa provides a mechanism to include data structured in RDF in HTML pages, providing a set of markup attributes that enable machine-readable information to be embedded in web pages. The formal specification is contained in the document 'RDFa Core 1.1 — Third Edition', while the approach is explained in the 'RDFa 1.1 Primer — Third Edition'. A dedicated website that outlines the objectives and the benefits of RDFa and provides pointers to informational materials and an index of tools is available at rdfa.info. (Source: http://www.w3.org/TR/rdfa-syntax)
RDF/XML	The RDF/XML serialisation is one of the possible ways of serialising RDF data, using an XML syntax (see also 'Turtle').
Revised legislation	We use the terms 'revise', 'revised' and 'revision' to refer to the editorial process of incorporating amendments and carrying through other effects into legislation.
Section	A provision, usually numbered, constituting the lowest level of division in the main body of an act or other primary legislation.
Serialisation	Serialisation is the process of translating data structures or object state into a format that can be stored (for example, in a file, or transmitted across a network connection link) and reconstructed later in the same or another computer environment.

Short title	The title by which an act or measure is usually known. It is to be distinguished from the long title, which sets out the purposes of the legislation.
Standard	A published specification for, e.g., the structure of a particular file format, recommended nomenclature to use in a particular domain, a common set of metadata fields, etc. Conforming to relevant standards greatly increases the value of published data by improving machine readability and easing data integration. (Source: http://opendatahandbook.org/glossary/en/terms/standard)
Statute	An item of primary legislation, such as an act or measure.
Statute book	A term we use to denote the totality of the statute law in force at any particular time.
Statutory instrument	A type of secondary legislation made under authority contained in acts of parliament.
SPARQL	SPARQL is a language to query RDF data. It is to RDF data what SQL is to relational databases. An overview of the main aspect of SPARQL is contained in the document 'SPARQL 1.1 Overview'. The formal specification is defined in the document 'SPARQL 1.1 Query Language'. (Source: Open Data Support)
Turtle	Turtle is one of the possible ways to serialise RDF data, using a text syntax. This is usually more compact and readable than RDF/XML.
URI	A uniform resource identifier (URI) is a compact sequence of characters that identifies an abstract or physical resource. URI enables interaction with representations of the resource over a network, typically the World Wide Web, using specific protocols. A URI can be composed of: URL: uniform resource locator URN: uniform resource name. The most common form of URI is the URL, but it can also be found as a URN or as a combination of both. This last option is the one that provides URI with more consistent characteristics for the identification of resources. (Source: ISA's 10 rules for persistent identifiers)
URL	Uniform resource locator. It can be defined as a URI that provides the means to access a web resource. URLs occur most commonly to reference web pages (http), but can also have a role in file transfer (ftp), email (mailto) or even database access (JDBC).
URN	Uniform resource name. It is the historical name for a uniform resource identifier (URI) that uses the urn scheme.
Version	1. 'Version' may refer to the 'as enacted' version of the legislation or the 'latest available (revised)' version. 2. 'Version' is also used in the context of revised legislation to refer to one of any number of versions of a provision (or higher level of division of legislation) that may exist in any number of different versions, usually created as a result of amendments made to it.

ANNEX II: ELI ontology

The latest version of the ELI ontology can be found on the Metadata Registry website [1].

[1] http://publications.europa.eu/mdr/eli/index.html

ANNEX III: Self-assessment check list

The following check list allows decision-makers to perform a self-assessment of their solution for publishing legislation, from an ELI perspective.

STEP	QUESTION	YES	NO	N/A
Organisation and policy				
1	Have you identified your business case? (1)	o	o	o
2	Have you estimated the resources for the implementation of the ELI pillars? (2)	o	o	o
3	Have you set up a governance structure? (3)	o	o	o
4	Have you set up an ELI implementation project? (4)	o	o	o
5	Have you formulated a URI and metadata policy? (5)	o	o	o
Pillar 1 — Identification of legislation				
1	Have you captured requirements relevant for HTTP URIs?	o	o	o
2	Have you designed URI templates? (6)	o	o	o
3	Have you configured the web server for ELI URI resolution? (7)	o	o	o
Pillar 2 — ELI metadata				
1	Have you captured requirements relevant for metadata? (8)	o	o	o
2	Have you designed the metadata schema? (9)	o	o	o
3	Have you developed the metadata publishing routines? (10)	o	o	o
4	Have you published the metadata? (11)	o	o	o
Pillar 3 — Render the ELI metadata machine-reusable				
1	Have you adapted the web pages to facilitate parsing and extraction?	o	o	o
2	Have you provided the metadata in a machine-readable format?	o	o	o

(1) Good practice 1: Building on the knowledge, experience and tools of others (support by the ELI Taskforce)

(2) Good practice 2: Estimating implementation costs

(3) Good practice 3: Setting up a central organisation as a national coordination point for ELI implementation

(4) Good practice 4: Structuring your implementation project

(5) Good practice 5: Writing a policy document providing guarantees for long-term persistence

(6) Good practices 6: Designing URI template(s) to stay as close as possible to existing citation practice; Good practice 7: Modelling HTTP URIs by treating each piece of legislation as a unique resource; Good practice 8: Taking into account the concepts of the ELI ontology: legal resource, legal expression and format; Good practice 9: Considering the right level of granularity when identifying legislation; Good practice 10: Following common good practices for HTTP URIs; Good practice 11: Testing URIs against real data.

(7) Good practice 12: Using content negotiation and/or identifiers for language and format variants; Good practice 13: Providing service-level guarantees; Good practice 14: Monitoring website usage and links to legislation on the web through statistics; Good practice 15: Monitoring service levels and persistence of URIs; Good practice 16: Following a user-centered design taking into account type of users

(8) Good practice 17: Considering the right level of granularity when creating metadata

(9) Good practice 18: Structuring information by linking it to other documents; Good practice 19: Creating metadata as part of the workflow, including validation services; Good practice 20: Generating standard metadata formats

(10) Good practice 21: Annotating legal information with RDFa markup

(11) Good practice 22: Dealing with legacy content

OA-04-17-130-EN-N

Publications Office

ISBN 978-92-78-41429-0